Asian Journal of Psychiatry 97 (2024) 104068

Contents lists available at ScienceDirect

Asian Journal of Psychiatry

journal homepage: www.elsevier.com/locate/ajp

Microbiota-brain axis: Exploring the role of gut microbiota in psychiatric disorders - A comprehensive review

Jawahar Singh [a,1], Vanlallawmzuali [b,1], Amit Singh [c], Suryanarayan Biswal [d], Ruth Zomuansangi [c], C. Lalbiaktluangi [c], Bhim Pratap Singh [e], Prashant Kumar Singh [f], Balachandar Vellingiri [g], Mahalaxmi Iyer [c], Heera Ram [h], Bharat Udey [a], Mukesh Kumar Yadav [c,*]

[a] Department of Psychiatry, All India Institute of Medical Sciences (AIIMS), Bathinda, Punjab, India
[b] Department of Biotechnology, Mizoram Central University, Pachhunga University College Campus, Aizawl, Mizoram, India
[c] Department of Microbiology Central University of Punjab, Bathinda 151401, India
[d] Department of Human Genetics and Molecular Medicine, Central University of Punjab, Bathinda 151401, India
[e] Department of Agriculture and Environmental Sciences (AES), National Institute of Food Technology Entrepreneurship and Management (NIFTEM), Sonepat, Haryana, India
[f] Department of Biotechnology, Pachhunga University College Campus, Mizoram University (A Central University), Aizawl 796001, Mizoram, India
[g] Stem cell and Regenerative Medicine/Translational Research, Department of Zoology, School of Basic Sciences, Central University of Punjab (CUPB), Bathinda, Punjab 151401, India
[h] Department of Zoology, Jai Narain Vyas University, Jodhpur, Rajasthan 342001, India

ARTICLE INFO

Keywords:
Gut microbiota
Mental health
Psychiatric disorder
Gut-brain axis
Secondary brain

ABSTRACT

Mental illness is a hidden epidemic in modern science that has gradually spread worldwide. According to estimates from the World Health Organization (WHO), approximately 10% of the world's population suffers from various mental diseases each year. Worldwide, financial and health burdens on society are increasing annually. Therefore, understanding the different factors that can influence mental illness is required to formulate novel and effective treatments and interventions to combat mental illness. Gut microbiota, consisting of diverse microbial communities residing in the gastrointestinal tract, exert profound effects on the central nervous system through the gut-brain axis. The gut-brain axis serves as a conduit for bidirectional communication between the two systems, enabling the gut microbiota to affect emotional and cognitive functions. Dysbiosis, or an imbalance in the gut microbiota, is associated with an increased susceptibility to mental health disorders and psychiatric illnesses. Gut microbiota is one of the most diverse and abundant groups of microbes that have been found to interact with the central nervous system and play important physiological functions in the human gut, thus greatly affecting the development of mental illnesses. The interaction between gut microbiota and mental health-related illnesses is a multifaceted and promising field of study. This review explores the mechanisms by which gut microbiota influences mental health, encompassing the modulation of neurotransmitter production, neuroinflammation, and integrity of the gut barrier. In addition, it emphasizes a thorough understanding of how the gut microbiome affects various psychiatric conditions.

1. Introduction

Humans are often referred to as "superorganisms" because of the presence of diverse microorganisms in their bodies. The number of microorganisms present in the human body is 1.3 times higher than the number of cells in the body, and each individual has approximately 38 trillion microorganisms (10^{12}) (Sender et al., 2016). Microorganisms reside in particular niches of the human body including the urogenital tract, respiratory tract, gastrointestinal tract, skin, and mammary glands. Interactions with these niches may be pathogenic, commensalism, or mutualistic (Ogunrinola et al., 2020). Among different niches, the human gut contains one of the most abundant and diverse groups of microbes, collectively known as the gut microbiome. Bacteria constitute approximately 99% of the gut microbiome, and 90% of the intestinal

* Corresponding author.
E-mail addresses: mukesh.yadav@cup.edu.in, mukiyadav@gmail.com (M.K. Yadav).
[1] First Authors Equal contribution

https://doi.org/10.1016/j.ajp.2024.104068
Received 22 November 2023; Received in revised form 28 February 2024; Accepted 1 May 2024
Available online 7 May 2024

flora of adults are dominated by *Firmicutes* and *Bacteroidetes* (Patrascu et al., 2017). Other phyla, such as *Proteobacteria*, *Firmicutes*, *Euryarchaeota*, *Actinobacteria*, and *Verrucomicrobia*, have also been identified (Fava et al., 2019). Table 1 shows the gut associated microbes and the recent findings on the factors affecting the gut microbiota composition. Technical advancements have revealed that gut microbiota plays an important role in the general health and well-being of humans. These microbes aid digestion, produce essential vitamins, and help maintain a healthy intestinal environment. They play a significant role in the immune system by influencing the ability to fight infections and defend against diseases. Some microorganisms in the gut are involved in the breakdown of complex carbohydrates and fibers, which the human body cannot digest on its own. This breakdown results in the production of short-chain fatty acids (SCFAs) and other metabolites that are not only a source of energy but also have protective effects on the gut lining. Microbes help to train the immune system and teach it to differentiate between harmful pathogens and beneficial microorganisms. A balanced gut microbiota is essential for the proper functioning of the immune system, and imbalances in gut microbiota are associated with various diseases. The composition of the gut microbiota can influence metabolism, weight control, and the risk of conditions, such as obesity and type 2 diabetes. Microbes can affect how the body processes nutrients and stores fats. Beneficial microorganisms in the body can help prevent harmful pathogens from gaining a foothold. They compete for resources, produce antimicrobial compounds, and maintain an environment inhospitable to potential invaders. They help in several physiological functions such as protection against foreign pathogens, digestion, weight gain or loss, immunity, growth, and regulation of host immunity (Alonso and Guarner, 2013; Collins et al., 2013; Piccioni et al., 2022). In contrast, dysbiosis of the gut microbiota has been associated with several diseases including cardiovascular disease, hypertension, anxiety, depression, cancer, obesity, inflammatory bowel disease, alcoholic liver disease, hepatocellular carcinoma, and diabetes (Afzaal et al., 2022; Morgan and Huttenhower, 2012; Pascal et al., 2018; Philips et al., 2022). Bidirectional communication between the central nervous system and gut microbiota has also been observed, which can cause several psychiatric illnesses, including anxiety, bipolar disorder (BD), stress, and depression (Morais et al., 2021). It is estimated that 970 million people worldwide suffer from mental health disorders (Dattani et al., 2023). In this review, the association between the gut microbiota and different psychiatric illnesses is presented in detail.

2. Gut microbiome – "the second brain"

Research on biological psychiatry has attracted attention because it has been observed that the human gut microbiota is linked to mental health and affects the brain and behavior (Crumeyrolle-Arias et al., 2014; Zheng et al., 2016). Therefore, it is crucial to be familiar with the gut microbiome and its linkage association with the brain to study mental health and different psychiatric disorders so that advanced treatments can be formulated.

2.1. The gut-brain axis

The human brain and microbes interact bidirectionally, which is known as the "MGB axis" (Malan-Muller et al., 2018). This bidirectional communication occurs through pathways such as the immune, endocrine, hypothalamic-pituitary hypothalamic-pituitary (HPA), limbic, metabolic, sympathetic afferent, afferent, efferent, and central nervous systems (Carabotti et al., 2015; Díaz-Anzaldúa et al., 2011; Mayer, 2011). In addition to these pathways, the integrity of the blood-brain barrier (BBB), HPA axis, intestinal epithelial barrier, and neurotransmitters produced by microbes, such as tryptophan and serotonin, contribute to MGB axis interaction (Braniste et al., 2014; O'Mahony et al., 2015). The MGB axis serves as the epicenter that enhances the signals of the brain and affects the sensory, secretory, and motor functions of the gut. At the same time, it also allows metabolites and signals from the gut microbiota to affect the biochemistry, behavior, development, and function of the brain (Cryan and O'Mahony, 2011; Marques et al., 2014). Changes in the accessibility of the blood-intestinal barrier can initiate innate immunity and enhance the inflammatory level of the cerebral system, thus resulting in vulnerability to various psychiatric disorders (Marano et al., 2023).

Recent studies have shown that important neurotransmitters are produced by the gut microbiota that controls the activity of the brain through immune pathways (Table 2) (Chu et al., 2019; Yano et al., 2015). For example, *Lactobacillus* and *Bifidobacterium* synthesize Gamma aminobutyric acid (GABA), which complements the inhibitory pathway in the brain (Patterson et al., 2019). *Oscillibacter* and *Lactobacillus* produce serotonin that promotes tryptophan synthase gene expression (Busnelli et al., 2019). Other bacteria such as *Lactobacillus plantarum, Enterococcus spp., Morganella morganii, Streptococcus spp., Klebsiella pneumoniae,* and *Escherichia spp.* are also capable of producing serotonin (Dinan et al., 2015). Reports have revealed that neurotransmitters and metabolites such as SCFAs, secondary bile acids, vitamins, and amino acids regulate various immune system pathways, which in turn affect memory, learning, locomotion, behaviours, and neurodegenerative disorders (Baj et al., 2019; Dalile et al., 2019). Anxiety, depression, and locomotor behaviour are influenced by the inflammasome pathway (Singhal et al., 2014).

2.2. Connection of gut-brain axis pathways to mental health

A previous study reported that rats induced with *Campylobacter jejuni* developed anxiety-like behaviours (Goehler et al., 2008). This is triggered by information produced from the gut along the vagus nerve, which activates neuronal brain regions (Goehler et al., 2005; Breit et al., 2018). The gut brain-axis network is reported to be bidirectional linking the enteric and the central nervous system not only anatomically but also includes humoral, immune, endocrine, and metabolic routes of communication (Appleton, 2018a). The response of the HPA axis to stress conditions was also observed by differentiating and comparing gnotobiotic, germ-free, and specific-pathogen-free mice. It was revealed that commensal microbes change the response exerted by the HPA axis under stress conditions.

2.2.1. Neurologic or neuroendocrine pathway

The neuroendocrine pathway comprises neurotransmitters, the enteric nervous system, and the vagus nerve. Approximately 90% of the massive networks of nerves within the muscles of the intestine are composed of the vagus nerve. Thus, the vagus nerve is a key component in the management of the gut-brain axis (Marano et al., 2023). The different nerves of the neuroendocrine pathways consist of the important hormone-like neurotransmitters that are produced by intestinal microbes such as serotonin (produced by *Enterococcus* spp., *Candida* spp., *Escherichia* spp.), histamine (produced by *Escherichia coli*, *Klebsiella aerogenes, Salmonella typhimurium*), GABA (produced by *Bifidobacterium* spp., *Lactobacillus* spp.), acetylcholine (*Lactobacillus* spp.), kynurenine (*Lactobacillus)*, and dopamine (*Enterococcus* spp., *Bacillus* spp.) (Góralczyk-Bińkowska et al., 2022). Gut bacteria interact with several cells in the gut endocrine system, including enteroendocrine cells (EECs). EECs contain signaling hormones, such as anorexigenic ([Neuropeptide Y (NPY) gene and Peptide YY (PYY) (NPY, PYY) and orexigenic (ghrelin), which function as neurotransmitters and target the vagus nerve (Osadchiy et al., 2011).

2.2.2. The hypothalamic–pituitary–adrenal (HPA) Axis

The hypothalamus consists of neuroendocrine cells that regulate cytokine secretion by releasing corticotrophin-releasing hormone (CRH) (Nezi et al., 2015). CRH then modulates the pituitary gland to activate adrenocorticotropic hormone (ACTH), which passes through the blood barrier (Allen and Sharma, 2018), leading to the activation of cortisol in

Table 1
Major factors affecting the gut microbiota composition, some of the associating gut microbes and recent research findings.

S. No.	Factors		Gut microbial taxa	Models	Findings (Examples)	References
1.	Mode of delivery	Vaginal	*Prevotella, Lactobacillus, Sneathia, Bacteroides, Escherichia, Bifidobacterium, Parabacteroides*	Women and their newborns	Infants delivered through the vagina obtain microbes that bore a resemblance to the vaginal microbes of the mother while those delivered through C-section attained microbes similar to the mother's skin. Vaginal birth acquires higher antibody responses against pneumococcal and meningococcal vaccines than C-section delivery.	(de Koff et al., 2022; Dominguez-Bello et al., 2010; Xiao and Zhao, 2023)
		Caesarian section	*Clostridia, Acinetobacter, Staphylococcus, Klebsiella, Enterobacter, Corynebacterium*			
2.	Diet		1. Prebiotics –Lactobacilli, ↑Bifidobacterium *Bacteroides, Clostridium ↓perfringens* 2. Animal proteins – *↑Lactobacillus, Bacteroidetes, Enterococcus, Lactococcus, Shigella flexneri, Parabacteroides, Lactobacillus murinus, Streptococcus hyointestinalis Roseburia, Akkermansia ↓muciniphila, Faecalibacterium prausnitzii, Ruminococcus bromii, Bifidobacterium animalis, Lactobacillus delbrueckii subsp. bulgaricus* 3. Vegetarian/Vegan diet – *↓Enterobacteriaceae*	Gut bacteria-*Escherichia coli* (NCTC10418) and *Enterococcus faecalis* (ATCC19433) and human intestinal epithelial cells-Caco-2 cells	Recent research on artificial sweeteners was conducted by demonstrating the influence of artificial sweeteners on the pathogenicity of gut bacteria and gut epithelium-microbiome communication. It was revealed that the artificial sweeteners enhanced biofilm formation of the gut bacteria and enhanced the gut bacteria to attach, infect and kill the human gut epithelial cells (host model).	(Liu et al., 2022; Amaretti et al., 2019; Shil and Chichger, 2021; Singh et al., 2017)
3.	Host genetics		*↑ Rikenellaceae, Prevotellaceae, Bacteroides, Faecalibacterium, Escherichia, Bifidobacterium, ↓Lachnospiraceae, ↓Bifidobacterium bifidum, Firmicutes, Clostridia*	Humans	In a Dutch Microbiome Project, 205 pathways and 207 taxa of gut microbiome in 7738 individuals were studied for the influence of host genes. The outstanding signal was observed at the lactate (LCT) locus, a gene accountable for the endurance of lactase in adults leading to the decrease in *Bifidobacterium longum* ($p = 3.2\times10{-}08$), *Actinobacteria* ($p= 2.3\times10{-}13$) and other higher taxa which is heritable.	(Lopera-Maya et al., 2022)
4.	Antibiotics		*↑ Enterobacteriaceae,Prevotella, Barnesiella, Alistipes Lactobacillus* spp., ↓Bacteroides spp., Bifidobacterium spp.,	Mice	Mice models were treated with antibiotics such as enrofloxacin, polymixin B sulfate and vancomycin. It was revealed the microbial composition of the mice's guts was different compared to controls. There was a decline in the species richness and diversity in the gut microbiome of mice treated with enrofloxacin and vancomycin. Additionally, the antibiotics also caused up-regulation of the gene including *IFN-γ, IL-1β, IL-6, Th17, IL-4, TNF-α* and *IL-23*.	(Sun et al., 2019)
5.	Socioeconomic	Urban	*↑ A. Putredinis, F. prausnitzii, Bacteroides, ↓ Prevotella, Adlercreutzia, Erysipelotrichaceae*	Humans	825 adults from different ethnic groups and races in the United States were observed for their gut microbiome diversity. It has been reported that lower socioeconomic status (SES) have diverse gut microbiota as compared to higher SES. *Prevotella copri* and the genus *Catenibacterum* were found to be the most abundant species in lower SES.	(Lapidot et al., 2022)
		Rural	↑ *Bifidobacterium* spp., *Faecalibacterium prausnitzii, Catenibacterium, Mitsuokella, Prevotella copri, Rothia muciliginosa, Eubacterium biforme, Sutterella, Alistipes putredinis, Dorea formigenerans,*			
6.	Age		*↑ Enterobacteriaceae, Clostridia, Odoribacter, Akkermansia, Butyricimonas, Christensenellaceae, Bilophila, Streptococcus, Eggerthella ↓ Lactobacillus, Bifidobacterium, Eubacterium rectale, Coprococcus,*	Newborns to centenarians	Results of raw 16 S rRNA sequences obtained from 368 samples showed that as the ages of individuals increase, the gut microbial diversity decreases while some microbes associated with cancer and inflammation increase.	(Chen et al., 2019; Ghosh et al., 2020; Odamaki et al., 2016; Xu et al., 2019)

(continued on next page)

Table 1 (*continued*)

S. No.	Factors	Gut microbial taxa	Models	Findings (Examples)	References
		Prevotella, Lachnospira, Faecalibacterium		Apart from the alteration in the gut microbiota, it has also been revealed that the synthesis of metabolites from the gut microbiome such as SCFAs has been substantially decreased which may be related to the process of aging.	

Table 2
Gut microbial composition and metabolites linked with different psychiatric disorders.

S. No.	Psychiatric disorders	Elevated Gut microbial taxa	Decreased gut microbial taxa	Neurotransmitters & metabolites	References
1.	Depression/Major Depressive Disorder (MDD)	*Oscillibacter, Anaerostipes, Phascolarctobacterium, Klebsiella, Blautia, Lachnospiraceae incertae sedis, Parasutterella, Clostridium, Atopobium, Parabacteroides, Streptococcus, Eggerthella*	*Dialister, Faecalibacterium, Bifidobacterium, Escherichia/Shigella, Ruminococcus*	Short chain fatty acids (SCFAs), catecholamine, Gamma-aminobutyric acid (GABA), histamine, tryptophan	Kim and Shin (2018); Zhu et al. (2022); Lowe et al. (2023)
2.	Bipolar disorder	*Coriobacteriaceae,* Flavonifactor, Betaproteobacterium, Clostridium Cluster IV, Atopobium cluster, *Lactobacillus,* Bifidobacterium, *Faecalibacterium prausnitzii,* Bacteroides–Prevotella group, Coriobacteria	*Clostridiaceae,* Bifidobacteria, Enterobacteriaceae, Roseburia, Ruminococcus, Coprococcus	L-glutamate, serotonin, GABA, tryptophan, dopamine, acetylcholine, norepinephrine.	Birner et al. (2017); Coello et al. (2019); Flowers et al. (2020); Morais et al. (2021); Ni et al. (2022); Ortega et al. (2023)
3.	Autism Spectrum Disorder (ASD)	*Anaerofilum, Clostridium* spp.*, Barnesiella intestinihominis, Dorea* spp.*,* Enterobacteriaceae, Clostridiales, *Pseudomonas, C. perfringens, B. intestinihominis, Turicibacter sanguinis, A. muciniphila, Burkholderia*	*Faecalibacterium* spp.*, Parasutterella excrementihominis, Turicibacter* spp.*, Roseburia* spp.*, Escherichia coli, Fusobacterium, Streptococcus, Oscillospira, Subdoligranulum, Lactobacillus, Staphylococcus*	Serotonin, Tryptophan such as indole-3-propionic acid (IPA), Glutamate, GABA, Noradrenaline, 5-hydroxyindoleacetic acid	De Angelis et al. (2013); Fowlie et al. (2018); Srikantha and Mohajeri (2019); Taniya et al. (2022)
4.	Attention-Deficit Hyperactivity Disorder (ADHD)	*Eubacterium hallii, Roseburia, Desulfovibrio, Butyricicoccus, Bifidobacterium, Peptostreptococcaceae, Ruminococcus torques, Oxalobacteraceae, B. uniformis, Collinsella*	*Clostridiales, Lachnospiraceae bacterium, Romboutsia, Lactobacillus, Bacteroides, B. coprocola, B. vulgatus, Parabacteroides, E. faecium, C. aerofaciens*	Noradrenaline, Dopamine, Choline, Serotonin, Xanthureic acid (XA), Quilonic acid (QA), Kynurenic acid (KA), SCFA	Aarts et al. (2017); Bull-Larsen and Mohajeri (2019); Chen et al. (2022); Liu et al. (2022); Fornaro et al. (2018); Kalenik et al. (2021)
5.	Obsessive-compulsive disorder (OCD)	*Peptostreptococcaceae, Ruminococcaceae, Rikenellaceae, Proteobacteria, Erysipelotrichaceae, Lachnospiraceae, Tenericutes, Deferribacteres, Clostridium butyricum*	*Anaerostipes, Odoribacter, Coprococcus, Oscillispira, Prevotellaceae, Agathobacter*	Glutamate (N-methyl- D-aspartate as a receptor and EAAT3 as a transporter), Serotonin (5-HT2AR), tryptophan Dopamine	Strandwitz et al. (2019); Domènech et al. (2022); Kong et al. (2022); Bendriss et al. (2023); Schiele et al. (2020)
6.	Schizophrenia	Anaerococcus, *Megasphaera, Klebsiella, Succinivibrio, Collinsella, Methanobrevibacter, Veillonellaceae, Coriobacteriaceae*	*Proteobacteria, Roseburia, Blautia, Haemophilus, Coprococcus, Sutterella, Norank, Enterobacteriaceae, Lachnospiraceae, Ruminococcus*	Glutamate (N-methyl- D-aspartate as a receptor, SCFAs, cysteine, methionine, Kynurenic acid, tryptophan	Nguyen et al. (2018); Shen et al. (2018); Zheng et al. (2019); Nguyen et al. (2021)
7.	Eating disorders (Anorexia nervosa and Bilumia nervosa)	*M. Smithii, Clostridia, Faecalibacterium prausnitzii, Enterobacteriaceae, Christensenellaceae, Lactococcus acidophilus,*	*Roseburia, Bifidobacterium, Haemophilus spp.,*	Isoleucine, tagatose, serotonin (receptor 1 A), rhamnose, xylose, sorbose, thionic acid, GABA, succinic acid, palmitic acid	Terry et al. (2022); Garcia and Gutierrez (2023); Fan et al. (2023); Agras (2001)

the adrenal gland (Hinds and Sanchez, 2022). Glucocorticoids, such as cortisol, direct homeostatic processes, repression, and control of T cells from overgrowth (Ramamoorthy and Cidlowski, 2016). Glucocorticoids have a remarkable effect on cognitive functions in the brain. Glucocorticoid levels influence brain function, particularly in the confinement of memory and learning (Tatomir et al., 2014). During chronic stress, an imbalance in hormone levels has been reported (Chu et al., 2021). For example, Wolkowitz et al. (2001) detected that under chronic stress the concentration of the hormone dehydroepiandrosterone was significantly decreased, whereas the concentration of the hormone cortisol increased or remained stable (Wolkowitz et al., 2001). The overproduction of glucocorticoids can result in the deterioration of hippocampal dendrites and neurons (Smedlund et al., 2021). Destruction of the hippocampal region can lead to post-traumatic stress disorder and depression (Bremner and Vermetten, 2012).

2.2.3. *Humoral or metabolic pathway*

The humoral pathway is regulated by metabolites produced by gut bacteria including SCFAs and lipopolysaccharides (LPS). SCFAs exhibit hormone-like activity and re-energizer-energized communication within nerve cells of the nervous system (He et al., 2020). SCFAs are produced in the gut, move across the blood-brain barrier to the brain, and help in mediating the homeostasis of the gut-brain axis (Silva et al., 2020). This process regulates the behaviour of human beings and is crucial for the proper development of the brain and the homeostasis of brain tissue (Frye et al., 2016). Serotonin produced by the gut resides in the plasma and stimulates sensory neurons that are directly linked to the central nervous system. Autism has been observed in children with elevated serotonin in the plasma (Marler et al., 2016). Recent studies also found a link between major depression and inflammatory response in which LPS and pro-inflammatory cytokines may induce symptoms of depression. This theory is confirmed by studies where higher levels of LPS are found in individuals with major depression than in controls (Mass et al., 2008).

As these LPS are pro-inflammatory, they induce inflammation in the gut or the intestine which increases intestinal permeability through which bacteria or their LPS can translocate to systemic circulation. This concept or hypothesis is termed as “leaky gut” (Fig. 1) which has been observed in diseases related to dysbiosis of gut microbiota like IBD (Fukui et al., 2016; Tulkens et al., 2020). Translocation of LPS of enterobacteria has also been found to play a role in the progression of inflammation in depression (Mass et al., 2008). The LPS were believed to be mostly produced endogenously and sustained by the gut microbiota rather than from external infection (Candelli et al., 2021). Perturbances in the gut can cause dysbiosis of the gut microbiota which can result in increased production of LPS in the gut. These endotoxins can disrupt the structure and function of important tight junction proteins like claudin-3 and zonulin eventually breaking down the tight junctions resulting in hyper-permeability of the intestinal epithelial cells (Moreno-Navarrete et al., 2012). The translocation of LPS can significantly affect the immune system in which production of TNF-α and IL-2 by antigen-presenting cells (APCs) and increased production of antibodies by β-lymphocytes were observed. The LPS then finally entered the bloodstream and caused acute or chronic endotoxemia (Candelli et al., 2021) (Fig. 1).

2.2.4. *Immune pathway*

An active connection exists between gut immune activation and neuroinflammatory processes in the brain (Sittipo et al., 2022). Intestinal microbes influence the inflammatory mechanisms of the gastrointestinal tract through cytokine production by the immune system (Rutsch et al., 2020). When gut dysbiosis occurs, there is an increase in peripheral inflammatory cytokines such as monocyte chemoattractant protein, IL-1β, CRP, TNF-α, IL-6, and IL-2 (Tang et al., 2020). These inflammatory cytokines are linked to neurodegenerative and psychiatric disorders such as depression, anxiety, and memory loss (Felger and Lotrich, 2013). These cytokines enter the central nervous system along the BBB or circumventricular organs and participate in microglial activation (Vakili et al., 2022). Moreover, neurotransmitters induced by gut bacteria, such as SCFAs, have been shown to directly influence the development of microglia in the gut (Dicks, 2022). This is consistent with a study performed on germ-free mice with a lower number of activated microglia compared to mice with a normal gut microbiome (Erny et al., 2015). Furthermore, unbalanced brain microglia, along with dysbiosis of the gut microbiome, have led to various mental health problems, such as anxiety, attention deficit hyperactivity disorder (ADHD), and depression (Fig. 2).

3. Impact of gut microbiota on mental health

Disruption of the bidirectional association between the neuroendocrine pathway and gut microbiota metabolites can lead to inflammatory responses that can manifest as psychiatric disorders, such as depression and irritable bowel syndrome. For example, increased production of pro-inflammatory cytokines like IL-1β, and IL-6 altered the metabolism of neurotransmitters by reducing the level of neurotransmitter precursors and via the activation of the HPA axis eventually contributing to the progression of depression. Moreover, it has been reported that the HPA axis, a major neuroendocrine system, can be activated by the gut microbiome, which regulates stress responses, mood, and emotions (Bao and Swaab, 2019). Studies have also revealed that hormones in the human gut play an important role in physiological processes that often result in different psychiatric disorders such as depression, stress, and anxiety (Sun et al., 2020). For example, there is a connection between mood disorders and obesity enhanced by various gut hormones of depression and anxiety such as ghrelin, cholecystokinin (CCK), and serotonin (5-HT) (Sun et al., 2020). Gut microbes can directly or indirectly synthesize neuroactive compounds called neurotransmitters including noradrenaline, GABA, histamine, dopamine, serotonin, tryptophan, melatonin, and SCFAs (Chen et al., 2021). The complexity of the gut microbial species and their genetic makeup enables the production of several novel and complex metabolites (Guinane and Cotter, 2013). Such metabolites act as a connection between the host and the gut microbiome, impacting the physiological and pathological processes of the host (Liu et al., 2022). Therefore, the exploration of these metabolites could aid in identifying possible targets for the treatment of psychiatric illnesses (Liu et al., 2023).

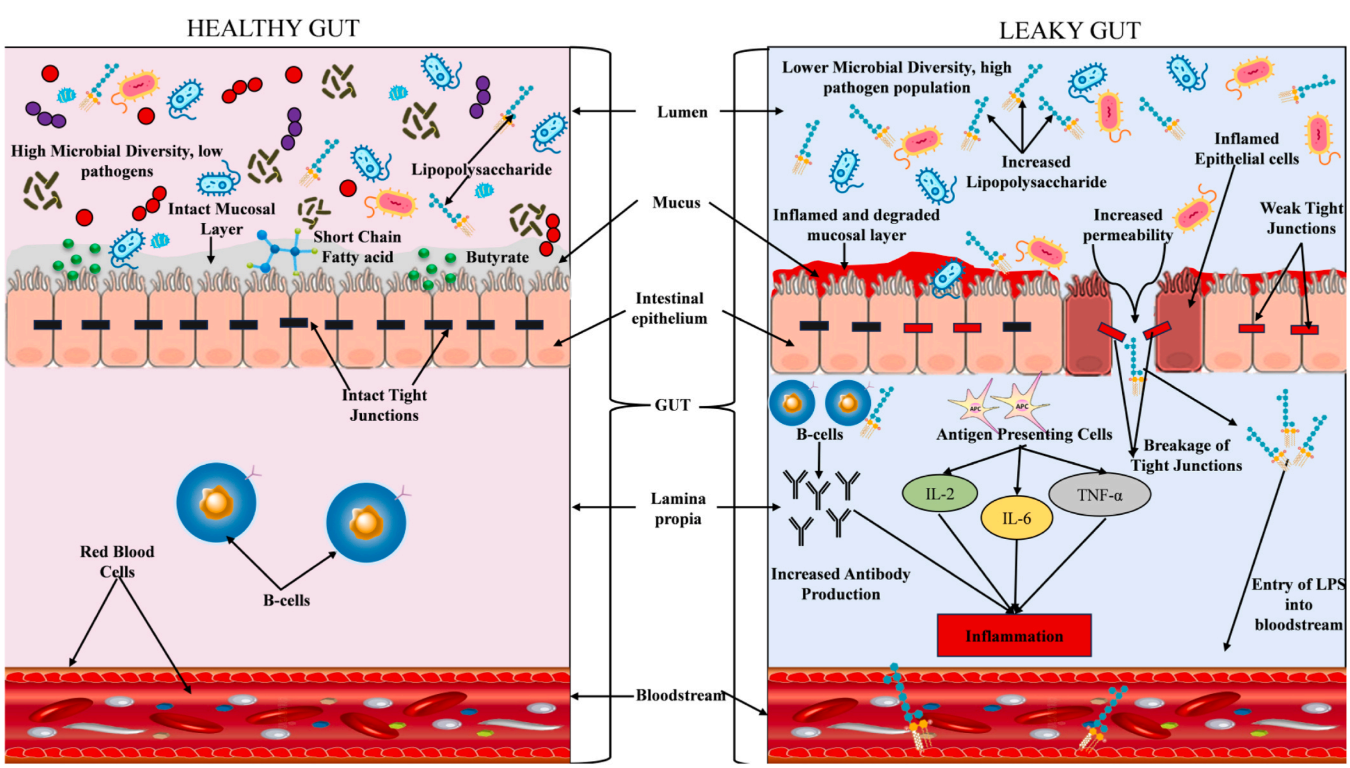

Fig. 1. A systematic representation of Leaky gut concept & mechanism and its relation with gut microbiota dysbiosis and disease.

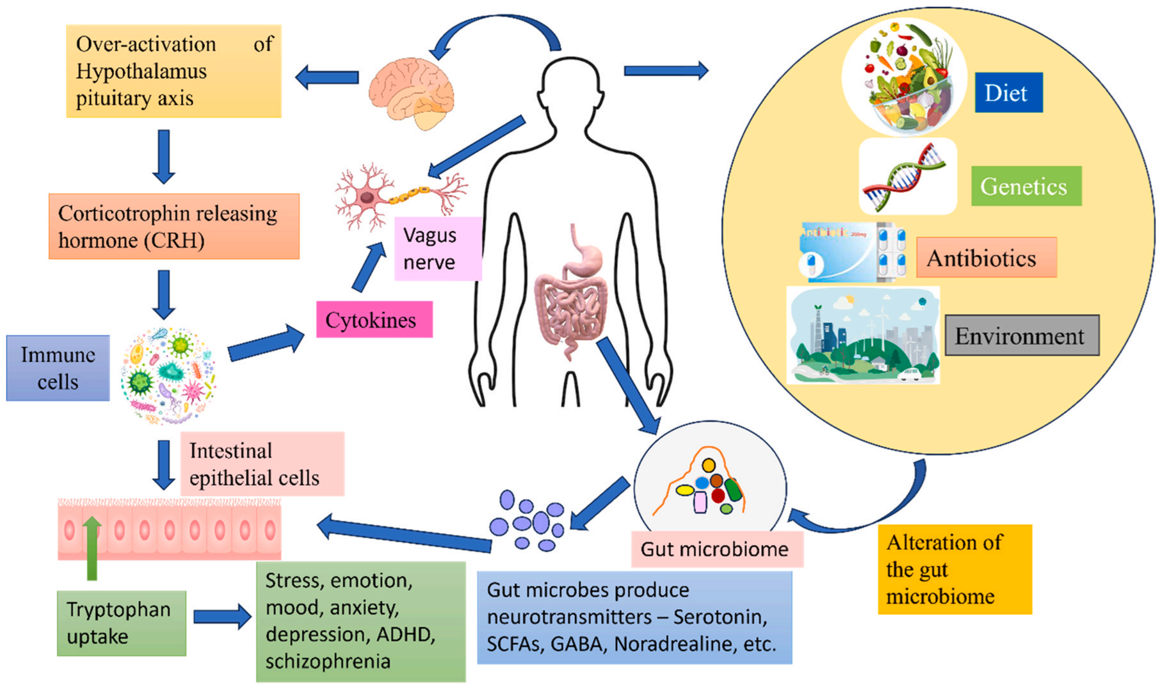

Fig. 2. Gut microbiota influence by various factors, connection of gut-brain axis and how the composition of gut microbiome influence mental health.

3.1. *Gut inflammation and different psychiatric disorders*

In recent years, there has been a significant increase in interest in elucidating the role of the gut microbiome in various common and severe mental disorders. Severe mental illness (SMI) (Schizophrenia, Bipolar disorder, etc.) is characterized by an increase in gut permeability (Nguyen et al., 2021). Enteric microbes are detected to exhibit systemic motion due to the debilitated lumen of the intestine (Canny and McCormick, 2008). The pro-inflammatory environment and other physiological malformations observed in SMI may be caused by dysbiosis of gut microbiota (Hsiao et al., 2013; Nguyen et al., 2021). Inflammation can cause intestinal permeability by disrupting barriers that protect the gut from pathogens (Di Tommaso et al., 2021). This phenomenon is also called "leaky gut," resulting in leakage of the blood-brain barrier (Morris et al., 2018).

3.1.1. *Stress*

Stress triggers the release of stress hormones such as cortisol, which can alter the composition and diversity of gut microorganisms. Alteration in the population of the gut microbiome due to stress can promote the development of chronic stress (Madison and Kiecolt-Glaser, 2019). Chronic stress is considered to have a negative impact on memory and may be connected with related to the composition of the gut microbiome (Kraimi et al., 2022). Germ-free mouse models inoculated with *Lactobacillus* species have been reported to exhibit enhanced short-term memory (Mao et al., 2020). In contrast, mice treated with antibiotics were observed to have detrimental effects on memory based on tests of novel object recognition and social transmission of food preference (Fröhlich et al., 2016). Similarly, a systematic review including 7 studies involving 443 subjects between the ages of 16–69 years found that *Bifidobacterium* spp. and *Lactobacillus* spp. decrease as a result of stress in humans (Anker-Ladefoged et al., 2021). Recently, microbes from chronically stressed individuals were transferred to germ-free Japanese quails via caecal microbiota transplantation (CMT). It was observed that the spatial learning memory of highly emotional and less emotional quails was disturbed by disrupting the proliferation of cells and disturbing the 3 weeks newborn cells in the hippocampus of the quails (Lormant et al., 2020). Under stress conditions, the permeability of the gut increases, leading to a decline in the activity of compact junction proteins and an increase in the transfer of large antigenic molecules (Fukui, 2016). However, it is uncertain whether the increase in leaky guts after stress is induced by altered gut microbiota or physiological factors (Clapp et al., 2017). Dysbiosis of the gut microbiota with a decline in *Bifidobacterium* and elevated lipopolysaccharide-producing bacteria has been reported to be linked to gut permeability (Leclercq et al., 2014). Understanding the intricate connections between stress, gut microbiota, and mental health is crucial for developing holistic approaches for the prevention and treatment of stress-related mental health problems.

3.1.2. *Depression*

Approximately 280 million people worldwide suffer from psychiatric disorders, and the mortality annually due to depression alone is estimated to be more than 700,000 (Institute of Health Metrics and Evaluation 2021; World Health Organization 2023).

Overactivation of the HPA axis and its mechanism can result in depression (Mikulska et al., 2021). Due to alterations in the gut microbiota, depressive mood and other neurobiological changes can lead to the production of LPS (Clapp et al., 2017). This triggered an inflammatory response. Microbial metabolites trigger the intestinal nerve, causing inflammation in the gastrointestinal tract, which, in turn, triggers the central nervous system (CNS). Neuroinflammation subsequently stimulates microglial activity and activates the kynurenine pathway (Cheung et al., 2019; Zhu et al., 2022). Symptoms that develop in the gut have been reported to be the most common symptom of depression (Dipnall et al., 2016). Patients with depression generally develop symptoms of gut-brain deterioration, such as metabolic disruption, appetite distraction, gastrointestinal disorders (irritable bowel syndrome), and abnormalities in the gut microbiome (Zhu et al., 2022). Studies have shown that there is an increase in the colonization of the phyla *Actinobacteria, Bacteroidetes,* and *Proteobacteria*, while *Firmicutes* decline in depression (Valles-Colomer et al., 2019; Zhang et al., 2022a; Zheng et al., 2016). Zhang et al. (2022a) experimented on mice and showed that the inflammatory response and permeability of the intestine are linked with gut microbiota dysbiosis (Zhang et al., 2022a). Due to endogenous melatonin reduction (EMR), there was a decrease in the genera *Lactobacillus* and *Bacteroidetes.* It was also reported that leaky gut and intestinal permeability were enhanced in EMR mouse models (Zhang et al., 2019). Several review studies have reported that *Desulfovibrio* and *Enterobacteriaceae* are higher while SCFA-producing microbes such as *Faecalibacterium* are lower in patients with depression than in controls (Simpson et al., 2021). A recent systematic review of 44 studies involving 2091 patients and 2792 controls found that depression is characterised by the increase of inflammatory bacteria like *Eggerthella, Flavonifractor,* and *Enterococcus* and a decrease of butyrate-producing bacteria like *Butyricicoccus, Faecalibacterium, Coprococcus, Fusicatenibacter*, etc. (Gao et al., 2023). Although more research is needed to fully elucidate the mechanisms at play, these findings open new avenues for potential interventions targeting the gut microbiota to alleviate and manage depression, highlighting the intricate interplay between gut health and emotional well-being.

3.1.3. Bipolar disorder

Similar to other psychiatric disorders, a reduction in gut microbial diversity has been observed in BD when compared with controls (Nikolova et al., 2021). The composition of gut microbiota is relatively important in identifying the phases, types, and severity of BD because the gut microbiota-produced metabolites can alter the brain function of the patient (Zhang et al., 2022b). For example, an analysis of the gut microbiome of 115 diagnosed bipolar patients found that patients with a higher *Faecalibacterium* population are associated with better health outcomes (Evans et al., 2017). Studies have found that the genera *Flavonifractor, Bacteroides–Prevotella* group, *Enterobacter spp., Faecalibacterium prausnitzii, Clostridium* Cluster IV, and *Atopobium* cluster were higher in patients with bipolar disorder, whereas the *Bifidobacteriobacter*-to-*Enterobacteriaceae* ratio was lower in bipolar disorder patients in compare to the healthy individuals (Aizawa et al., 2019; Coello et al., 2019; Ni et al., 2022).

Host–microbiota communication generally occurs in the mucosa of the intestine (Thursby and Juge, 2017). The microbiota-gut-brain axis has an impact on the physio-pathological emergence of bipolar disorder by regulating the CNS and enteric system to exert action on the inflammatory processes of the system, intestine, and brain and by producing important metabolites such as GABA, monoamines, acetylcholine, and glutamate (Ortega et al., 2023; Fornaro et al., 2018). In the kynurenic pathway, tryptophan 2,3-dioxygenase *(TDO2)* (hepatically) and indoleamine 2,3-dioxygenase 1 (IDO1) (extra-hepatically) decompose dietary tryptophan into kynurenine (Zhang et al., 2022a). Kynurenine is then broken down into metabolites, such as kynurenic acid (KynA), which is then metabolized into a neurotoxin called quinolic acid (Benedetti et al., 2020; Zhang et al., 2022a). Overexpression of KynA results in BD (Ortega et al., 2023; Zhang et al., 2022a).

Recently, the diversity of the gut microbiota of 23 patients with BD was compared to that of 23 healthy individuals (McIntyre et al., 2021). Microbial diversity decreased among BD patients with an abundance of *Collinsella* and *Clostridiaceae* compared to healthy subjects (McIntyre et al., 2021). A systematic review of 44 studies involving 2510 patients and 2407 controls found the low level butyrate-producing bacteria in patients with bipolar disorder (McGuinness et al., 2022). Probably, the decreased population of butyrate-producing bacteria results in decreased butyrate production, which may foster inflammation in bipolar patients (Sublette et al., 2021). In addition, the mood swings in bipolar disorder are found to be associated with certain cytokines such as interleukin-1β (IL-1β), tumor necrosis factor-α (TNF-α), and interleukin-6 (IL-6) (Munkholm et al., 2013; O'Brien et al., 2006; Ortiz-Dominguez et al., 2005)

3.1.4. Autism spectrum disorder (ASD)

Nearly 83%-91% of ASD cases are associated with gastrointestinal disorders (Vellingiri et al., 2022). ASD has been reported to cause disturbances in sleep, self-harm, aggressive behavior, and tantrums, thus worsening the behavior of children with autism and gastrointestinal problems (Iovene et al., 2017). Gut bacterial species, such as *Staphylococcus*, *Streptococcus,* and *Enterococcus*, are transferred from the mother to the autistic fetus. Later, the diversity of gut microbes increased, and the families were abundant *Bacteroidetes* and *Firmicutes*. The etiology of ASD has been suggested to be linked to the abundance of *Clostridia* spp. in patients with autism. Metabolites associated with *Clostridium* spp., such as 3-hydroxy hippuric acid, 3-(3-hydroxyphenyl)-3-hydroxy propionic acid, and 3-hydroxyphenyl acetic acid, have been found in children with autism, disrupting the metabolism of phenylalanine (Li and Zhou, 2016).

Studies have revealed that spore-forming *Clostridia* species can cause damage to thalamic axon outgrowth by decreasing gene expression in the brain, such as NTNG1, and minimizing thalamocortical axonogenesis (netrin-G1a+ thalamocortical axons) (Yaguchi et al., 2014). A recent study also reported that mothers with a higher abundance of *Bacteroidetes* may considerably increase netrin-Gla+ thalamocortical axons (Vuong et al., 2020). Vuong et al. (2020) also found that when pathogen-free (SP), antibiotic-treated (ABX), and germ-free (GF) mice were subjected to thalamic explants, there was a significant increase in specific metabolites, including N, N, N-trimethyl-5- amino valerate (TMAV), 3-indoxyl sulfate (3-IS), trimethylamine-N-oxide (TMAO), hippurate (HIP), and imidazole propionate (IP) in the offspring of SP mice compared to the offspring of ABX mice and GF mice. Forssberg (2019) conducted a study by separating infant rats every day for 3 hours from 2 to 15 days of the postnatal period, which results in dysbiosis of the gut. Due to chronic stress, there was an increase in *Clostridium* and *Bacteroides* as well as chemokines (CCL2 and IL-6) and pro-inflammatory cytokines (Forssberg, 2019).

3.1.5. Attention-Deficit Hyperactivity Disorder (ADHD)

Since there is currently no effective treatment, psychological and behavioral therapies are commonly used for ADHD (Schleupner and Carmichael, 2022). Microbiota transfer therapy (MTT) and fecal microbiota transplantation (FMT) are potential techniques for treating ADHD (Dash et al., 2022). A recent study was conducted on 6–10-year-old children (23 male ADHD patients, 7 female ADHD patients, and 30 control subjects) in a hospital in Taiwan (Wang et al., 2020). Individuals with ADHD have a more diverse gut microbiota composition than controls (Wang et al., 2020). ADHD individuals had an abundance of *Fusobacterium* while controls had higher levels of *Lactobacillus* (Wang et al., 2020). There were consequential changes in the composition of certain bacterial species between ADHD subjects and controls such as *Bacteroides coprocola*, *Bacteroides ovatus*, *Bacteroides uniformis,* and *Sutterella stercoricanis*. These bacteria could be the potential biomarkers of ADHD (Wang et al., 2020). Therefore, the authors concluded that dysbiosis of the genus may contribute to ADHD by disturbing the function and development of brain *Bacteroides* (Wang et al., 2020). However, in a more recent study, compared to controls there was so significant difference in the alpha and beta diversity of gut microbiota among 100 ADHD patients of which 49% were females (Richarte et al., 2021).

3.1.6. Obsessive-compulsive disorder (OCD)

The exact cause of OCD is unclear till today. However, several research studies point towards the potential cause of the disorder. The causes of OCD may be due to disturbances in the neurobiological function, environmental influences, and genetic susceptibility (Chen et al., 2023). Studies have reported that OCD can be inherited through genes such as the dopamine D2 receptor (DRD2) encoding gene and the serotonin transporter gene (SLC6A4) (Mattheisen et al., 2015).

A study on the stool samples of 22 OCD patients revealed that individuals with OCD had a lower diversity of gut microbes as compared to the control group and a considerably lower abundance of the genera *Anaerostipes, Oscillospira,* and *Odoribacter* (Turna et al., 2020). These bacteria produce SCFAs that can prevent the occurrence of a 'leaky gut' thus, preventing inflammation in the gut (Zhang et al., 2019). The study has revealed that the bacterial abundance reduction results in lower production of SCFAs which could lead to deterioration of health. In a recent study of gut dysbiosis and dysbiosis of the oropharyngeal microbiome of individuals with OCD, it was observed that the *Rikenellaceae* family that is linked to the gut inflammatory process increases while the genus *Coprococcus* decreases (Domènech et al., 2022). Decreased α-diversity has also been observed in OCD patients in several studies (Axelsson et al., 2019; Jiang et al., 2018; Troyer et al., 2021).

3.1.7. Schizophrenia

Neurotransmitters and metabolites contribute to the association of gut microbiota and schizophrenia. Gut inflammatory processes and the cause of schizophrenia have been reported to be associated, in such a way that one of the peptides in the host called 'alarmins' initiates signaling pathways that result in the development of several inflammatory diseases (Ma et al., 2022). Alteration of the gut microbiome composition and gut disruption leading to systemic inflammation by inflammasome activation is considered to be connected with schizophrenia and the regulation of tryptophan-kynurenine metabolism and neurotransmission of glutamatergic mechanism.

Gut microbes from schizophrenia patients were transferred to mice models and it was found that there was a distinct reduction in the tryptophan catabolism of the serotonin pathway while there was an increase in the Kynurenic pathway (Zhu et al., 2020). This Kyn-Kyna pathway is associated with schizophrenia through the regulation of the tryptophan-kynurenine mechanism (Zhu et al., 2020). It was also found that mice models receiving gut microbiome of schizophrenia also showed changes in lipids and amino acids as well as an interruption in the cycle of glutamate–glutamine–GABA, thus, decreasing glutamate in the brain (Zheng et al., 2019).

A recent study showed that there is no notable distinction in the composition of alpha-diversity, while there was a notable decline in the composition of beta-diversity in the gut microbiome of schizophrenia patients (Nocera and Nasrallah, 2022). Previous studies have reported a decline of *Proteobacteria* in schizophrenia patients compared to controls, while another study has observed an abundance in beta-diversity with an increase in *Fusobacteria* and *Proteobacteria* and a decrease in *Firmicutes* (Nguyen et al., 2018; Shen et al., 2018). Several studies have also confirmed that the diversity of the gut microbiome in schizophrenia patients changes on treatment with antipsychotic drugs (Ma et al., 2020; Schwarz et al., 2018). For example, schizophrenia individuals showing the first episode of psychosis were treated with risperidone and it was revealed that there was considerable alteration in the gut bacteria, which was considered to be linked with metabolic alteration induced by antipsychotic medications (Yuan et al., 2018). Schizophrenia patients without medication were found to have an abundance of alpha and beta-diversity than control subjects (Zhu et al., 2020).

3.1.8. Eating disorders (Anorexia nervosa and bulimia nervosa)

Strict dietary limitation in eating disorders causes detained gastric emptying and delayed transit time leading to bloating and satiety, thus, promoting restrictive behaviour through psychological and physiological mechanisms (Terry et al., 2022). These processes are linked to the distinct profile of gut microbial diversity. Furthermore, delayed transit time causes constipation, which in turn leads to an increase in the production of SCFAs in the gut microbiota (Tian et al., 2021). The melanocortin system (MC) takes part in the processes of inflammation, sexual function, homeostasis, and pigmentation (Gantz and Fong, 2003). In the case of anorexia nervosa, elevation in the activity of the MC system leads to dysfunction of neurotransmitter communication such as dopamine and serotonin (Steiger, 2004). This change in the neurotransmitter functioning disturbs the normal feeding and behaviour of anorexia nervosa individuals (Micioni Di Bonaventura et al., 2020). Avoidant/restrictive food intake disorder (ARFID) is also an associated psychiatric disorder of ED caused due to loss of interest, sensory susceptibility, and anxiety and is also regulated by the same gut microbiota of ED patients (Seetharaman and Fields, 2020).

Anorexia nervosa (AN) patients were found to have an abundance of Christensenellaceae family and β-diversity of bacteria at the genus level. Species such as *Blautia* sp., *Roseburia inulinivorans, Enterocloster bolteae, Erysipelatoclostridium ramosum, Roseburia intestinalis,* and butyrate-producing bacteria were also reported to inhabit the gut at high levels (Fan et al., 2023). A recent systematic study of 9 studies revealed that changes in the gut microbiota in AN patients may be distinguished by a reduction in species that produce butyrate and an increase in species that break down mucus (Di Lodovico et al., 2021). *C. paraputrificum* was identified as a key bacterium in causing thinness by producing several tryptophan metabolites such as indole acrylic acid, tryptamine, IPA, and indoleacetic acid which all take part in the modulation of mental health and appetite (Roager and Licht, 2018). SCFAs can lower the appetite by increasing the glutamatergic transfer of pro-opiomelanocortin (POMC) and cocaine- and amphetamine-regulated transcript (CART) pathways and GABAergic pathway (Chambers et al., 2015). SCFA also decreases gut permeability by promoting the expression of tight junction metabolites such as zonulin, occluding, and claudin-5 (He et al., 2020).

4. *In-vitro* and *In-vivo* analysis

The alteration of various gut microbiota was recorded in studies and the metabolites of these microbiota are recorded to cause various alterations regarding both improving and worsening of neurodegenerative disorders (Santoro et al., 2018). Table 3 shows the population of gut microbes in patients with neurodegenerative disorders. Specific focus was made on PD, in a study done by Scheperjans et al. (2015). They observed downregulation and upregulation of the phylum *Actinobacteria, Firmicutes,* and *Bacteroidetes* respectively, and they did not observe any changes in any specific species (Scheperjans et al., 2015). The upregulation of the *Bacteroidetes* phylum has seemed to play a critical role in the case of PD (Keshavarzian et al., 2015; Scheperjans et al., 2015; Zapała et al., 2021). 16sRNA is one of the major techniques that have been used in the detection of population change in the microbiota of various human samples (Clarridge, 2004; Johnson et al., 2019). Among the mentioned studies Lin et al. (2018) have observed more upregulation in the families of *Streptococcaceae, Eubacteriaceae, Lachnospiraceae, Aerococcaceae, Methylobacteriaceae, Comamonadaceae, Halmonadaceae, Hyphomonadaceae, Brucellaceae, Xanthomonadaceae, Sphingomonadaceae, Pasteurellaceae, Desulfovibrionaceae, Bifidobacteriaceae, Actinomycetaceae, Micrococcaceae, Intrasporangiacea*, and *Brevibacteriaceae*. The detection was done by using an equal number of controls from the sample hence the accurate changes in the microbiota changes were recorded (Lin et al., 2018). The studies conducted on human samples that compared the increased and decreased gut populations are listed below (Lin et al., 2018).

4.1. Ongoing clinical trials

The observation of gut dysbiosis in various diseases has led to the in-depth study of gut microbes and approaches therapeutically by modulating gut microbiota (Peterson et al., 2015). Dietary regulation has been considered a primary therapeutic approach and can be achieved by compounds such as prebiotics, probiotics, and symbiotics (Peterson et al., 2015). Only a limited number of clinical studies have been carried out in terms of prebiotic, probiotic treatment, and the treatment investigating the Gut-brain axis (Brüssow, 2019; Mörkl et al., 2020). Fermented milk-containing strains (*Acetobacter* sp*., A. aceti, L. fermentum, Lactobacillus delbrueckii delbrueckii, E. faecium, L. fructivornas, Leuconstoc spp., Candida krusei and L. Kefiranofaciens*) was given to 13 individuals with Alzheimer's disease (AD) and observed an increase in markers of oxidative stress, systemic inflammation and blood cell damage and overall cognitive score improvement (Ton et al., 2020). In the trial of 79 AD patients fed with probiotics – *Bifidobacterium longum, Lactobacillus acidophilus,* and *Bifidobacterium bifidum* along with 200 μg selenium or

Table 3
Population of gut microbes in patients with neurodegenerative disorders.

Neurodegenerative disease	Type of patients	Changes in the population								Techniques used	Reference
		Phylum		Family		Genera		Species			
		↑	↓	↑	↓	↑	↓	↑	↓		
Parkinson's disease (PD)	Human fecal sample (n=72: PD, n=72: Control)	Bacteroidetes	Actinobacteria, Firmicutes	*Gemellaceae, Bacteroidaceae, Rikenellaceae*	*Ruminococcaceae, Turicibacteraceae, Peptostreptococcaceae, Clostridiaceae, Mogibacteriaceae, Bifidobacteriaceae*	*BLlsutia, Phascolarctobacterium, Gemella, Bacteroides, Alistipes, Bilophila*	SMB53, *Dialister, Clostridium, Turicibacter,* cc115, *Bifidobacterium, Aldlercreutzia*			16 SrRNA Sequencing	Scheperjans et al. (2015)
	Human fecal sample (n=35: Control, n=45: PD)				*Enterobacteriaceae*	*Lactobacillus*	*C. leptum, Pseudomonas*		*Bacteroides fragilis, Clostridium coccoides, C. perfringens*	16S or 23 S rRNA RT-PCR	Hasegawa et al. (2015)
	n=28: controls (only male), n=31: PD (only male)	Bacteroidetes, Proteobacteria, Verucomicrobia	*Firmicutes*	*Bacteroidaceae, Clostridiaceae, Verrucomicrobiaceae*	*Lachnospiraceae, Coprobacillaceae*	*Akkermansia, OscilloSpira, Bacteroide*	*BlauTia, Coprococcus, Roseburia, Dorea*	Akkermansia muciniphila, Alistipes shahii	*Prevotella copri, Eubacterium biforme. Clostridium saccharolyticum*	16SrRNA Sequencing	Keshavarzian et al. (2015)
	Human fecal sample (n=14: control, n=24: PD)	Actinobacteria, Proteobacteria	Bacteroidetes	*Enterobacteriaceae, Veillonellaceae, Erysipeotrichaceae, Coriobacteriaceae, Streptococcaceae, Moraxellaceae, Enterococcaeceae*	*Prevotellaceae, Ruminococcus, Lachnospiraceae*	*Acidaminococcus, Acinetobacter, Enterococcus, Escherichia-Shigella, Megamonas, Megasphaera, Proteus, Streptococcus*	*Blautia, Faecalibacterium, Ruminococcus*			16SrRNA Sequencing	Li et al. (2017)
	Human fecal sample (n=34: controls, n=34: PD)		Bacteroidetes	*Enterobacteriaceae*	*Prevotellaceae, Enterococcacear, Lactobacillaceae*			*Akkermansia mucinipjila*	*Faecalibacterium prausnitzii*	RT-qPCR	Unger et al. (2016)
Alzhiemer's Disease (AD)	Human fecal sample (n=10: controls, n=40: AD)							*Escherichia/ Shigella*	*Eubacterium rectale, Bacteroides fragilis*	RT-qPCR	Cattaneo et al. (2017)
	Human fecal sample (n=31: control, n=34: AD)	Actinobacteria	Bacteroidetes, Verrucomicrobia	*Ruminococcaceae*	*Lanchnospiraceae, Bateroidaceae, Veillonellaceae*	*Dorea, Lactobacillus, Streptococcus, Bifidobacterium, Blautia, Escherichia*	*Alistipes, Bacteroides, Parabacteroides, Sutterella, Paraprevotella*			16SrRNA Sequencing	Zhuang et al. (2018)

(*continued on next page*)

Table 3 (*continued*)

Neurodegenerative disease	Type of patients	Changes in the population								Techniques used	Reference
		Phylum		Family		Genera		Species			
		↑	↓	↑	↓	↑	↓	↑	↓		
Multiple Sclerosis (MS)	Human fecal sample (n=17: controls, n=18: MS)			*Bifidobacterium, Desulfovibrio, Christensenellaceae*	*Lachnospiraceae, Ruminococcaceae*	*Akkermansia, Parabacteroides, Methanobrevibacter, Bacteroides*				16SrRNA Sequencing	Tremlett et al. (2016)
	Human fecal sample (n=31: controls, n=30: MS)							*C. perfringens* type B	*C. perfringens* type A	RT-qPCR	Rumah et al. (2013)
	Human fecal sample (n=40: controls, n=20: MS)					*Bifidobacterium, Streptococcus*	*Bacteroides, Faecalibacterium, Prevotella, Anaerostipes*			16SrRNA Sequencing	Miyake et al. (2015)
	Human fecal sample (n=36: controls, n=31: MS)				*Erysipelotrichaceae, Lachnospiraceae, Veillonellaceae*	*Pedobacter, Flavobacterium, Blautia, Dorea, Pseudomonas, Mycoplana*	*Adlercreutzia, Collinsella, Parabacteroides, Lactobacillus, Coprobacillus, Haemophilus*			16SrRNA Sequencing	Chen et al., (2016)

placebo for 12 weeks, when observed for metabolic parameters and cognition there was a significant increase in capacity of antioxidant, glutathione and scores of Mini-Mental State Examination (MMSE) and reduced levels of triglycerides, high-sensitivity C-reactive protein (hs-CRP) serum, total cholesterol and low-density lipoprotein (LDL) and levels of insulin (Tamtaji et al., 2019). A controlled clinical trial done with 30 AD individuals whose diet consists of multi-strain probiotics comprising *L. casei, L. acidophilus, L. fermentum*, and *B. bifidum* showed an increase in Beta cell function, MMSE scores, and a decrease in triglycerides, levels of malondialdehyde, insulin resistance and CRP (Akbari et al., 2016). The AD part of the trials was covered to have a comparison with Parkinson's disease (PD). Tamtaji et al. (2019) observed down-regulated levels of insulin, MDS-UPDRS, insulin resistance and low high-sensitivity CPR and monoaldehyde and upregulated levels of glutathione in a clinical trial with 60 PD patients aimed at observing metabolic restrictions and movement, given with multi-strain probiotics which comprises of Bifidobacterium bifidum, placebo, *Lactobacillus acidophilus, L.reuteri* for 12 weeks (Tamtaji et al., 2019). A clinical trial aimed at the observation of therapeutic effects of probiotics in 50 Parkinson's patients by the assessment of Inflammation, insulin and lipid profiles, an upregulation of PBMCs PPAR-γ and TGF-β and inflammatory IL-8, IL-1 and TNF-α downregulation was observed due to an altered diet, comprises of *Bifidobacterium bifidum*, placebo, *Lactobacillus acidophilus, L. reuteri* – a probiotic packed with multi-strains (Borzabadi et al., 2018). Since there was a downregulation in the inflammatory interleukins, it is said to have a neuroprotective effect (Borzabadi et al., 2018).

In Ayurvedic medicine *Mucuna pruriens* is the natural source of levodopa, the seed contains L-Dopa (Lampariello et al., 2012). Various alkaloids and trace amounts of 5-hydroxytryptamine and been used for PD treatment (Katzenschlager et al., 2004). Recently, two clinical trials on PD patients using 15 or 30 g and 200 mg of *M. pruriens* were done in 8 and 2 PD patients respectively (Katzenschlager et al., 2004). 30 g of *M. pruriens* had a faster effect onset and shorter latencies in *M. pruriens* and downregulated L-Dopa bioavailability after M. pruriens administration respectively (Contin et al., 2019; Katzenschlager et al., 2004) (Table 4).

5. Future therapeutics using gut microbiota

5.1. Probiotics, prebiotics, and synbiotics

The widely recognized two-way neuroendocrine system between the CNS and GI tract is the gut-brain axis, which plays an essential part in response to stress (Appleton, 2018b). Modification of the gut-brain axis has an association with neurological disorders (Mayer, 2011). The modification in the homeostasis of gut microbiota is considered a new therapeutic strategy for CNS and gastrointestinal-driven disorders and one way to achieve this was by the dietary ingestion of prebiotics and probiotics (Di Meo et al., 2018).

5.1.1. Probiotics

Probiotics are live organisms that can deliver a health benefit for the host when an adequate amount is administered (Di Meo et al., 2018). *Bacteroidetes* dominate fiber-rich food, food-consuming individuals' flora, and also have SCFAs which act as anti-inflammatory mediators (Portincasa et al., 2022). Furthermore, SCFAs also help to employ an effect of neuromodulators such as provoking GLP-1 (Glucagon-like peptide-1), leptin, and Ghrelin hormonal synthesis which in turn protect cells from toxicity induced by Aβ oligomers by acting as a neurotrophic factor (Dalile et al., 2019; Ho et al., 2019; Martin et al., 2013). The immunomodulation of the host is associated with the effect on health by probiotics for instance, the Interleukin (IL) IL-12/IL-10 ratio can be stabilized by *Bifidobacterium infantis*. Only a minimal study has been undertaken to explore the treatment of brain diseases by targeting gut microbiota. One clinical trial study has found that healthy patients are less prone to psychological disorders when treated with *Lactobacillus helveticus* and *Bifidobacterium*, indicating that it exerts a neuroprotective effect (Messaoudi et al., 2011). Similarly, another study suggested that the use of probiotics in milk can improve cognitive health (Benton et al., 2007). A study carried out by Magistrelli et al. (2019) with probiotics on PD revealed that a decrease in the pro-inflammatory cytokines results from *Lactobacillus salivarius* and *Lactobacillus acidophilus* and an increase in anti-inflammatory (Magistrelli et al., 2019).

5.1.2. Prebiotics

Dietary components that can alter beneficiary microorganism's

Table 4
Gut microbiota related diseases and the treatment mediated via gut bacteria/metabolites.

S. No	Subjects	Diet/drug used	Placebo treatment	Interval	Effects observed when compared to placebo	Reference
1.	Alzheimer's disease patients n=13	Strains of Acetobacter sp., A. aceti, L. fermentum, Lactobacillus delbrueckii delbrueckii, E. faecium, L. fructivornas, Leuconstoc spp., Candida krusei and L. Kefiranofaciens in a fermented milk.	×	12 weeks	↑markers of oxidative stress, systemic inflammation and blood cell damage and overall cognitive score improvement	(Ton et al., 2020)
2.	Alzheimer's disease patients n=79	Multistrain probiotic comprising B. bifidum, L. acidophilus and B. longum along with 200 ug selenium	✓	12 weeks	Compared to placebo therapy, treatment of selenium with probiotic have shown ↑glutathione, total antioxidant capacity and MMSE scores ↓triglycerdes, hs-CRP serum, total cholesterol LDL and also the insulin levels	(Tamtaji et al., 2019)
3.	Alzheimer's disease patients n=30	Multi-strain probiotic comprising L. casei, L. acidophilus, L. fermentum and B. bifidum	×	12 weeks	↑Beta cell function, MMSE scores ↓ Triglycerides, levels of malondialdehyde, insulin resistance and CRP	(Akbari et al., 2016)
4.	Parkinson's disease patients n=60	Multi-strain probiotic comprising B. bifidum, L. fermentum, L. acidophilus, L. reuteri	×	12 weeks	↓Malondialdehyde, high sensitive CRP, insulin, MDS-UPDRS and insulin resistance ↑Glutathione	(Tamtaji et al., 2019)
5.	Parkinson's disease patients n=50	Multi-strain probiotic comprising L. fermentum, B. bifidum, L. fermentum, L. acidophilus, L. reuteri	✓	12 weeks	↓ Inflammatory IL-8, TNF-α and IL-1 ↑PBMCs PPAR-γ and TGF-β	(Borzabadi et al., 2018)
6.	Parkinson's disease patients n=8	Mucuna pruriens (15 g or 30 g) or L-Dopa	×	1 dose	Faster effect onset and shorter latencies in M. pruriens	(Katzenschlager et al., 2004)
7.	Parkinson's disease patients n=2	M. pruriens (200 mg)	×	2 doses	↓ L-Dopa bioavailability after M. pruriens administration	(Contin et al., 2015)

activity are called Prebiotics (Di Meo et al., 2018). These are utilized by microorganisms as a substrate and produce metabolites, micronutrients which will be used by the host. Prebiotics also induce the growth of species that are selectively beneficial especially *Bifidobacterium* and *Lactobacilli* hence the prebiotics can regulate brain health (Di Meo et al., 2018). Studies that have dealt with prebiotics in both animals and human shows positive effects and also showed an increase in BDNF, a key neurotrophin associated with neuronal growth and survival (Petschow et al., 2013; Sudo et al., 2004). Hence prebiotics and probiotics can alter the chemistry of the brain by increasing the BDNF expression via activity of the gut hormones.

5.1.3. Synbiotics

A combination of prebiotics and probiotics is symbiotic, an improvement of GI symptoms was observed in a cohort of PD in a randomized controlled trial (RCT) using a synbiotic strain (Barichella et al., 2016). A 20.7% and 34.3% decrease in the incidence of enteritis and VAP in sepsis patients was observed when a synbiotic mixture of the probiotic *Bifidobacterium breve* and the prebiotic galacto-oligosaccharide was administered (Shimizu et al., 2018). A 3-month clinical trial with 20 obese patients was carried out using a synbiotic which consisted of the prebiotic galacto-oligosaccharide mixture and a probiotic *Bifidobacterium lactis, Lactobacillus acidophilus, Bifidobacterium longum* and *Bifidobacterium bifidum.* Although they did not detect any statistically significant change in the body mass, body fat, BMI, etc of symbiotic and placebo groups, they observed that the symbiotic increased the population of butyrate-producing bacteria like *Lactobacillus* and *Bifidobacterium* (Sergeev et al., 2020). Synbiotics supplementation in milk formula fed to 290 healthy infants ranging from 6 to 19 weeks was reported to increase the *Bifidobacteria* population and decrease the abundance of *C. difficile* (Phavichitr et al., 2021).

5.2. Polyphenols

It was suggested by several studies that neuroprotection can be achieved by polyphenols such as Curcumin, Epigallocatechin-3-gallate (EGCG), Resveratrol, etc., which can modulate cellular functions (Di Meo et al., 2018). Curcumin is said to protect the integrity of BBB and it also has a neuroprotective function in cerebral ischemia according to several *in vivo* and *in vitro* studies (Fan and Lei, 2022; Jiang et al., 2018; Khayatan et al., 2022). By binding to the amyloid fibers curcumin inhibits amyloid beta aggregation and oligo formation, and the same effect goes on alpha-synuclein in the case of PD (Singh et al., 2013). Hence, curcumin exerts an anti-aggregative effect on both AD and PD (Yang et al., 2022). EGCG has been proven to protect against age-related cognitive decline and protects amyloid precursor protein-producing enzymes (Mandel et al., 2004; Sutherland et al., 2005). In a study done on rodents, EGCG was able to achieve a reduction in Alzheimer's cerebral amyloidosis and an improvement in learning ability and spatial cognition (Rezai-Zadeh et al., 2005). Resveratrol has been noted for its protection against neurodegenerative diseases and synaptic membranal oxidative stress reduction (Sun et al., 2020). By preventing the beta fibrils formation, and extension, and by destabilization of the formed ones, amyloid beta-induced toxicity can also be prevented by Resveratrol (Ono et al., 2006). Overall, the alteration of metabolites of microbiota can be a novel therapeutic approach towards neuropathologies, bioavailability of biomolecules such as polyphenols and their available metabolites in the systemic circulation have been highlighted in studies and said to increase and decrease protective signaling molecules and stress signaling, inflammatory/oxidative capacity respectively (Manach et al., 2004).

5.3. Fecal microbiota transplantation (FMT)

The transferring of feces from donors that are normal, healthy, and have gut microbiota to unhealthy patients via nasointestinal, enema nasogastric or endoscopic approaches to re-establish the normal flora of microbiota and to re-establish the functionality of the existing microbiota (Parker et al., 2022; Staley et al., 2017). Since there is a bidirectional signal interface through the gut-brain axis, FMT is considered to be a potential treatment for neurodegenerative diseases (Gulati et al., 2020). Transplantation of fecal from AD mice to normal mice showed deprived cognitive function and minimal nervous system-associated fecal metabolites such as Valine, GABA, and taurine (Fujii et al., 2019). Previous studies revealed that the fecal microbiota transplantation to antibiotic-treated (Amyloid precursor protein) APP/(Presenilin) PS1–21 transgenic mice from APP/PS1–21 male transgenic mice revealed the restoration of the usual microbiome of the intestine and also partly reversed the morphology of microglia and pathology of Aβ (Dodiya et al., 2019). Nevertheless, the treatment of FMT is still in the stage of infancy, and studies have been undertaken to evaluate the effectiveness of FMT in neurological diseases. Trials are undergone in human models for the identification of effective and potential transplants e.g., Parasites, bacteria, viruses, etc (Woodworth et al., 2017). The ideal composition of microbiota to be transplanted is still under examination and the human trials are at the initial stage and raise safety apprehensions.

6. Current approaches and future prospectives

Many approaches can be employed for the treatment of mental illnesses. The fecal microbiota transplant is one of the most employed approaches for gut microbiota dysbiosis alignment (Biazzo and Deidda, 2022). The transfer of microbes through the feces of a healthy person to the recipient (patient or model) helps in the study of the connection of gut microbiome by observing the physiological, psychological, and behavioral aspects and conducts positive modification and restoration of the gut microbiome (Gupta et al., 2016). However, FMT needs to be considered as it could transfer immune-regulatory bacteria as well as pathogenic bacteria, and the determination of desired microbes could be imprecise (Ortega et al., 2023). Moreover, the benefits and prolonged effects of this technique are still uncertain. Improvement and development of this technique are crucial as it is an important therapeutic approach.

Immune-based strategies including anti-inflammatory and non-steroidal drugs such as fluoxetine, celecoxib, sertraline, and reboxetine are common treatments for psychiatric illnesses available today (Fitton et al., 2022; Müller, 2017). Novel and safe treatments have focused on receptors or signaling pathways and these are highly required for the treatment of individuals with MDD along with antidepressants (Murrough et al., 2017). Additionally, antibiotics such as minocycline, doxycycline, etc., are known for having anti-inflammatory, anti-depressant, and neuroprotective activities ref. Minocycline particularly reduces the composition of gut bacteria such as Bacteroidetes and Firmicutes while hindering the expression of COX-2 and decreases the levels of prostaglandins E2 that plays an important role in the microglial activation, thus, leading to the amelioration of psychotic and depressive symptoms (Miyaoka et al., 2012).

Probiotics especially *Bifidobacteria* and *Lactobacillus* serve as a promising therapy for the treatment of psychological disorders (Johnson et al., 2021). They enhance the function of the mucosal barrier, thus rejuvenating the leaking of the gut. However, it is still controversial as the stability, consistency, and composition of probiotics vary greatly in different strains with uncertainty in the duration, dosage, and resistance from the host immune system. Prebiotics promote positive alteration in the gut microbiota by enhancing the growth of some favorable gut microbes. Therefore, diet contributes to a great extent in managing the gut microbiota which in turn maintains the gut-brain axis. Light therapy and social rhythm are also used as good mood balancers and favorable changes in the gut microbiota have also been reported (Steardo et al., 2020). Therefore, vigorous and authentic humanized rodent models need to be developed (Fig. 3).

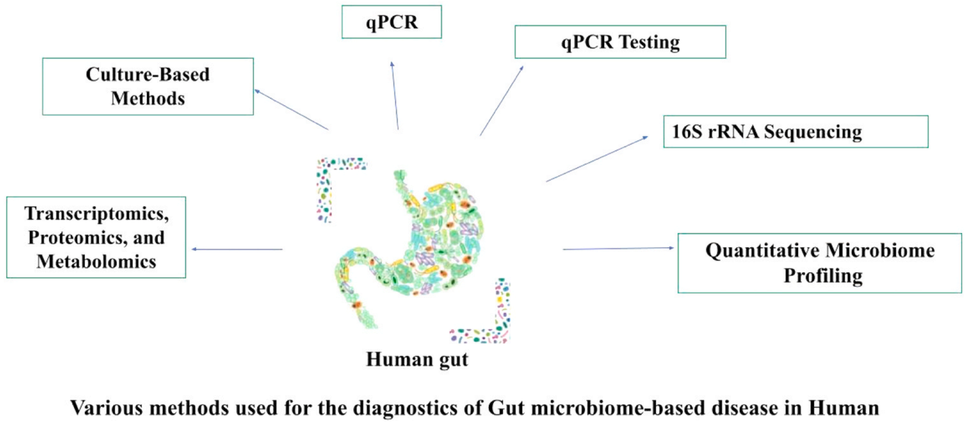

Fig. 3. Numerous diagnostic approaches used in gut microbiome.

Three-dimensional (3D) models of the human gut and brain along with novel co-culture systems such as the vagus nerve could be an excellent alternative for further study of gut-brain axis mechanisms (Han and Jang, 2022; Moysidou and Owens, 2021). More studies on metagenomic, metaproteomic, and metabolomic should be done to obtain proper and better data. Advanced computational programs are now available that can provide next-level accuracy on the results especially in population-based studies, for example, populations at risk of brain disorders (Putignani et al., 2019). By using the technique of bioinformatics, accurate results and data could be obtained for a better understanding of clinical studies. Advancement in new tools and software could be an alternative in discovering the early onset psychotic symptoms, earlier recognition of high-risk individuals, and identifying the immunological and microbial dysbiosis or modifications so that earlier treatment could be accomplished (Arnold et al., 2016; Young et al., 2021).

7. Limitations

This review included studies with cross-sectional data however, the limitation of this review is that we do not take into account the temporal variations of the gut microbiome and the difference in the gut microbial composition depending on different geographic regions. In addition, this review focused mostly on the bacteria community of the gut microbiome and does not included studies on a myriad of other microbes like viruses, fungi and bacteriophages that also inhabit the GI tract. Lastly, the bacterial composition studies were primarily carried out using 16 S sequencing which may not be completely accurate due to its limited resolution, lower sensitivity, and misinformation of non-culturable microbes (Poretsky et al., 2014). However, the use of omics techniques like meta-transcriptomics, metagenomics, and metabolomics will greatly improve the exploration of gut microbial composition.

8. Conclusion

The evidence gathered from numerous studies supports that the intricate connections between the gut and the brain offer new avenues for preventive and therapeutic strategies, with the potential to enhance mental health and overall well-being. A significant finding indicates the function of the gut microbiota in regulating neurotransmitter production and its connection with the brain leading to deterioration of mental health. By nurturing a healthy gut microbiota through lifestyle modifications, diet maintenance, and targeted interventions, we can potentially unravel a new frontier for improving mental well-being and ease the problems of mental health problems on a global scale. This could have a significant socioeconomic impact on any country like India, where the global disease burden of psychiatric disorders is high. The gut-brain axis, once viewed as a fascinating hypothesis, is now a substantial and encouraging avenue for ameliorating mental health outcomes in the future. Further research and clinical trials are warranted to elucidate the full extent of the gut-brain axis's impact and to refine microbiota-based interventions.

Declaration of Competing Interest

None to declare

Acknowledgement

This study is supported by funding to MKY from the Science and Engineering Research Board (SERB), Govt. of India, vide Project Sanction No: SRG/2021/000371, and RSM grant sanction (CUPB/Director (R&D)/2023/133) by Central University of Punjab, Bathinda, Punjab India.

Financial Disclosure

None

References

Aarts, E., Ederveen, T.H., Naaijen, J., Zwiers, M.P., Boekhorst, J., Timmerman, H.M., Arias Vasquez, A., 2017. Gut microbiome in ADHD and its relation to neural reward anticipation. PloS One 12 (9), e0183509.
Afzaal, M., Saeed, F., Shah, Y.A., Hussain, M., Rabail, R., Socol, C.T., Hassoun, A., Pateiro, M., Lorenzo, J.M., Rusu, A.V., 2022. Human gut microbiota in health and disease: Unveiling the relationship. Front. Microbiol. 13, 999001.
Agras, W.S., 2001. The consequences and costs of the eating disorders. Psychiatr. Clin. North Am. 24, 371–379.
Aizawa, E., Tsuji, H., Asahara, T., Takahashi, T., Teraishi, T., Yoshida, S., Koga, N., Hattori, K., Ota, M., Kunugi, H., 2019. Bifidobacterium and Lactobacillus counts in the gut microbiota of patients with bipolar disorder and healthy controls. Front. Psychiatry 9, 730.
Akbari, E., Asemi, Z., Daneshvar Kakhaki, R., Bahmani, F., Kouchaki, E., Tamtaji, O.R., Hamidi, G.A., Salami, M., 2016. Effect of Probiotic Supplementation on Cognitive Function and Metabolic Status in Alzheimer's Disease: A Randomized, Double-Blind and Controlled Trial. Front Aging Neurosci. 8, 256. https://doi.org/10.3389/fnagi.2016.00256.
Allen, M.J., Sharma, S., 2018. Physiology, adrenocorticotropic hormone (ACTH).
Alonso, V.R., Guarner, F., 2013. Linking the gut microbiota to human health. Br. J. Nutr. 109, S21–S26.
Amaretti, A., Gozzoli, C., Simone, M., Raimondi, S., Righini, L., Pérez-Brocal, V., Rossi, M., 2019. Profiling of protein degraders in cultures of human gut microbiota. Frontiers in microbiology 10, 2614.
Anker-Ladefoged, C., Langkamp, T., Mueller-Alcazar, A., 2021. The potential impact of selected bacterial strains on the stress response. : Healthc. MDPI 494.
Appleton, J., 2018a. The gut-brain axis: influence of microbiota on mood and mental health. Integr. Med (Encinitas) 17, 28–32.
Appleton, J., 2018b. The gut-brain axis: Influence of microbiota on mood and mental health. Integr. Med.: A Clin. 'S. J. 17, 28.
Arnold, J.W., Roach, J., Azcarate-Peril, M.A., 2016. Emerging technologies for gut microbiome research. Trends Microbiol. 24, 887–901.

Axelsson, P.B., Clausen, T.D., Petersen, A.H., Hageman, I., Pinborg, A., Kessing, L.V., Bergholt, T., Rasmussen, S.C., Keiding, N., Løkkegaard, E.C.L., 2019. Relation between infant microbiota and autism?: results from a national cohort sibling design study. Epidemiology 30, 52–60.

Baj, A., Moro, E., Bistoletti, M., Orlandi, V., Crema, F., Giaroni, C., 2019. Glutamatergic signaling along the microbiota-gut-brain axis. Int. J. Mol. Sci. 20, 1482.

Bao, A.-M., Swaab, D.F., 2019. The human hypothalamus in mood disorders: the HPA axis in the center. IBRO Rep. 6, 45–53.

Barichella, M., Pacchetti, C., Bolliri, C., Cassani, E., Iorio, L., Pusani, C., Pinelli, G., Privitera, G., Cesari, I., Faierman, S.A., Caccialanza, R., Pezzoli, G., Cereda, E., 2016. Probiotics and prebiotic fiber for constipation associated with Parkinson disease: An RCT. Neurology 87, 1274–1280. https://doi.org/10.1212/WNL.0000000000003127.

Bendriss, G., MacDonald, R., & McVeigh, C., 2023. Microbial Reprogramming: A Promising Approach for the Treatment of Obsessive-Compulsive Disorders.

Benedetti, F., Aggio, V., Pratesi, M.L., Greco, G., Furlan, R., 2020. Neuroinflammation in bipolar depression. Front. Psychiatry 11, 71.

Benton, D., Williams, C., Brown, A., 2007. Impact of consuming a milk drink containing a probiotic on mood and cognition. Eur. J. Clin. Nutr. 61, 355–361. https://doi.org/10.1038/sj.ejcn.1602546.

Biazzo, M., Deidda, G., 2022. Fecal microbiota transplantation as new therapeutic avenue for human diseases. J. Clin. Med. 11, 4119. https://doi.org/10.3390/jcm11144119.

Birner, A., Platzer, M., Bengesser, S.A., Dalkner, N., Fellendorf, F.T., Queissner, R., Reininghaus, E.Z., 2017. Increased breakdown of kynurenine towards its neurotoxic branch in bipolar disorder. PloS One 12 (2), e0172699.

Borzabadi, S., Oryan, S., Eidi, A., Aghadavod, E., Daneshvar Kakhaki, R., Tamtaji, O.R., Taghizadeh, M., Asemi, Z., 2018. The effects of probiotic supplementation on gene expression related to inflammation, insulin and lipid in patients with Parkinson's disease: a randomized, double-blind, placebocontrolled trial. Arch. Iran. Med. 21, 289–295.

Braniste, V., Al-Asmakh, M., Kowal, C., Anuar, F., Abbaspour, A., Tóth, M., Korecka, A., Bakocevic, N., Ng, L.G., Kundu, P., 2014. The gut microbiota influences blood-brain barrier permeability in mice. Sci. Transl. Med. 6, 263ra158-263ra158.

Breit, S., Kupferberg, A., Rogler, G., Hasler, G., 2018. Vagus nerve as modulator of the brain–gut axis in psychiatric and inflammatory disorders. Front. Psychiatry 44.

Bremner, J.D., Vermetten, E., 2012. The hippocampus and post-traumatic disorders. The clinical neurobiology of the hippocampus: an integrative view. Oxford University Press, Oxford, pp. 262–272.

Brüssow, H., 2019. Probiotics and prebiotics in clinical tests: an update. F1000Res 8, 1157. https://doi.org/10.12688/f1000research.19043.1.

Bull-Larsen, S., Mohajeri, M.H., 2019. The potential influence of the bacterial microbiome on the development and progression of ADHD. Nutrients 11 (11), 2805.

Busnelli, M., Manzini, S., Chiesa, G., 2019. The gut microbiota affects host pathophysiology as an endocrine organ: a focus on cardiovascular disease. Nutrients 12, 79.

Candelli, M., Franza, L., Pignataro, G., Ojetti, V., Covino, M., Piccioni, A., Franceschi, F., 2021. Interaction between lipopolysaccharide and gut microbiota in inflammatory bowel diseases. Int. J. Mol. Sci. 22 (12), 6242.

Canny, G.O., McCormick, B.A., 2008. Bacteria in the intestine, helpful residents or enemies from within? Infect. Immun. 76, 3360–3373.

Carabotti, M., Scirocco, A., Maselli, M.A., Severi, C., 2015. The gut-brain axis: interactions between enteric microbiota, central and enteric nervous systems. Ann. Gastroenterol.: Q. Publ. Hell. Soc. Gastroenterol. 28, 203.

Cattaneo, A., Cattane, N., Galluzzi, S., Provasi, S., Lopizzo, N., Festari, C., Ferrari, C., Guerra, U.P., Paghera, B., Muscio, C., Bianchetti, A., Volta, G.D., Turla, M., Cotelli, M.S., Gennuso, M., Prelle, A., Zanetti, O., Lussignoli, G., Mirabile, D., Bellandi, D., INDIA-FBP Group, 2017. Association of brain amyloidosis with pro-inflammatory gut bacterial taxa and peripheral inflammation markers in cognitively impaired elderly. Neurobiol. Aging 49, 60–68.

Chambers, E.S., Morrison, D.J., Frost, G., 2015. Control of appetite and energy intake by SCFA: what are the potential underlying mechanisms? Proc. Nutr. Soc. 74, 328–336.

Chen, L.L., Abbaspour, A., Aspvall, K., Rück, C., Bulik, C.M., Pascal, D., 2023. Longitudinal study of gut microbiome in obsessive–compulsive disorder. Brain Behav., e3115

Chen, J., Chia, N., Kalari, K.R., Yao, J.Z., Novotna, M., Paz Soldan, M.M., Luckey, D.H., Marietta, E.V., Jeraldo, P.R., Chen, X., Weinshenker, B.G., Rodriguez, M., Kantarci, O.H., Nelson, H., Murray, J.A., Mangalam, A.K., 2016. Multiple sclerosis patients have a distinct gut microbiota compared to healthy controls. Sci. Rep. 6, 28484.

Chen, Y.C., Lin, H.Y., Chien, Y., Tung, Y.H., Ni, Y.H., Gau, S.S.F., 2022. Altered gut microbiota correlates with behavioral problems but not gastrointestinal symptoms in individuals with autism. Brain, Behav., Immun. 106, 161–178.

Chen, Yijing, Xu, J., Chen, Yu, 2021. Regulation of neurotransmitters by the gut microbiota and effects on cognition in neurological disorders. Nutrients 13, 2099.

Chen, R., Xu, Y., Wu, P., Zhou, H., Lasanajak, Y., Fang, Y., Zhao, J., 2019. Transplantation of fecal microbiota rich in short chain fatty acids and butyric acid treat cerebral ischemic stroke by regulating gut microbiota. Pharmacological Research 148, 104403.

Cheung, S.G., Goldenthal, A.R., Uhlemann, A.-C., Mann, J.J., Miller, J.M., Sublette, M.E., 2019. Systematic review of gut microbiota and major depression. Front. Psychiatry 10, 34.

Chu, B., Marwaha, K., Sanvictores, T., Ayers, D., 2021. Physiology, stress reaction, in: StatPearls [Internet]. StatPearls Publishing.

Chu, C., Murdock, M.H., Jing, D., Won, T.H., Chung, H., Kressel, A.M., Tsaava, T., Addorisio, M.E., Putzel, G.G., Zhou, L., 2019. The microbiota regulate neuronal function and fear extinction learning. Nature 574, 543–548.

Clapp, M., Aurora, N., Herrera, L., Bhatia, M., Wilen, E., Wakefield, S., 2017. Gut microbiota's effect on mental health: The gut-brain axis. Clin. Pract. 7, 987.

Clarridge, J.E., 2004. Impact of 16S rRNA Gene Sequence Analysis for Identification of Bacteria on Clinical Microbiology and Infectious Diseases. Clin. Microbiol Rev. 17, 840–862. https://doi.org/10.1128/CMR.17.4.840-862.2004.

Coello, K., Hansen, T.H., Sørensen, N., Munkholm, K., Kessing, L.V., Pedersen, O., Vinberg, M., 2019. Gut microbiota composition in patients with newly diagnosed bipolar disorder and their unaffected first-degree relatives. Brain, Behav., Immun. 75, 112–118.

Collins, S.M., Kassam, Z., Bercik, P., 2013. The adoptive transfer of behavioral phenotype via the intestinal microbiota: experimental evidence and clinical implications. Curr. Opin. Microbiol. 16, 240–245.

Contin, M., Lopane, G., Mohamed, S., Calandra-Buonaura, G., Capellari, S., De Massis, P., Nassetti, S., Perrone, A., Riva, R., Sambati, L., Scaglione, C., Cortelli, P., 2019. Clinical pharmacokinetics of pramipexole, ropinirole and rotigotine in patients with Parkinson's disease. Park. Relat. Disord. 61, 111–117. https://doi.org/10.1016/j.parkreldis.2018.11.007.

Contin, M., Lopane, G., Passini, A., Poli, F., Iannello, C., Guarino, M., 2015. Mucuna pruriens in Parkinson Disease: A Kinetic-Dynamic Comparison With Levodopa Standard Formulations. Clin Neuropharmacol 38, 201–203.

Crumeyrolle-Arias, M., Jaglin, M., Bruneau, A., Vancassel, S., Cardona, A., Daugé, V., Naudon, L., Rabot, S., 2014. Absence of the gut microbiota enhances anxiety-like behavior and neuroendocrine response to acute stress in rats. Psychoneuroendocrinology 42, 207–217.

Cryan, J.F., O'Mahony, S.M., 2011. The microbiome-gut-brain axis: from bowel to behavior. Neurogastroenterol. Motil. 23, 187–192.

Dalile, B., Van Oudenhove, L., Vervliet, B., Verbeke, K., 2019. The role of short-chain fatty acids in microbiota–gut–brain communication. Nat. Rev. Gastroenterol. Hepatol. 16, 461–478.

Dash, S., Syed, Y.A., Khan, M.R., 2022. Understanding the role of the gut microbiome in brain development and its association with neurodevelopmental psychiatric disorders. Front. Cell Dev. Biol. 10, 880544.

Dattani, S., Rodés-Guirao, L., Ritchie, H., Roser, M., Ortiz-Ospina, E., 2023. Suicides. Our World in Data.

De Angelis, M., Piccolo, M., Vannini, L., Siragusa, S., De Giacomo, A., Serrazzanetti, D.I., Francavilla, R., 2013. Fecal microbiota and metabolome of children with autism and pervasive developmental disorder not otherwise specified. PloS One 8 (10), e76993.

de Koff, van Baarle, D., van Houten, Reyman, M., Berbers, G.A., van den Ham, F., et al., 2022. Mode of delivery modulates the intestinal microbiota and impacts the response to vaccination. Nature Communications 13 (1), 6638.

Di Lodovico, L., Mondot, S., Doré, J., Mack, I., Hanachi, M., Gorwood, P., 2021. Anorexia nervosa and gut microbiota: A systematic review and quantitative synthesis of pooled microbiological data. Prog. Neuropsychopharmacol. Biol. Psychiatry 106, 110114. https://doi.org/10.1016/j.pnpbp.2020.110114.

Di Meo, F., Donato, S., Di Pardo, A., Maglione, V., Filosa, S., Crispi, S., 2018. New Therapeutic Drugs from Bioactive Natural Molecules: The Role of Gut Microbiota Metabolism in Neurodegenerative Diseases. Curr. Drug Metab. 19, 478–489. https://doi.org/10.2174/1389200219666180404094147.

Di Tommaso, N., Gasbarrini, A., Ponziani, F.R., 2021. Intestinal Barrier in Human Health and Disease. Int J Environ Res Public Health 18, 12836.

Díaz-Anzaldúa, A., Díaz-Martínez, A., Díaz-Martínez, L.R., 2011. The complex interplay of genetics, epigenetics, and environment in the predisposition to alcohol dependence. Salud Ment. 34, 157–166.

Dicks, L.M., 2022. Gut bacteria and neurotransmitters. Microorganisms 10, 1838.

Dinan, T.G., Stilling, R.M., Stanton, C., Cryan, J.F., 2015. Collective unconscious: how gut microbes shape human behavior. J. Psychiatr. Res. 63, 1–9.

Dipnall, J.F., Pasco, J.A., Berk, M., Williams, L.J., Dodd, S., Jacka, F.N., Meyer, D., 2016. Into the bowels of depression: unravelling medical symptoms associated with depression by applying machine-learning techniques to a community based population sample. PloS One 11, e0167055.

Dodiya, H.B., Kuntz, T., Shaik, S.M., Baufeld, C., Leibowitz, J., Zhang, Xulun, Gottel, N., Zhang, Xiaoqiong, Butovsky, O., Gilbert, J.A., Sisodia, S.S., 2019. Sex-specific effects of microbiome perturbations on cerebral Aβ amyloidosis and microglia phenotypes. J. Exp. Med 216, 1542–1560. https://doi.org/10.1084/jem.20182386.

Domènech, L., Willis, J., Alemany-Navarro, M., Morell, M., Real, E., Escaramís, G., Bertolín, S., Sánchez Chinchilla, D., Balcells, S., Segalàs, C., Estivill, X., Menchón, J. M., Gabaldón, T., Alonso, P., Rabionet, R., 2022. Changes in the stool and oropharyngeal microbiome in obsessive-compulsive disorder. Sci. Rep. 12, 1448. https://doi.org/10.1038/s41598-022-05480-9.

Dominguez-Bello, M.G., Costello, E.K., Contreras, M., Magris, M., Hidalgo, G., Fierer, N., Knight, R., 2010. Delivery mode shapes the acquisition and structure of the initial microbiota across multiple body habitats in newborns. Proceedings of the National Academy of Sciences 107 (26), 11971–11975.

Erny, D., Hrabě de Angelis, A.L., Jaitin, D., Wieghofer, P., Staszewski, O., David, E., Keren-Shaul, H., Mahlakoiv, T., Jakobshagen, K., Buch, T., 2015. Host microbiota constantly control maturation and function of microglia in the CNS. Nat. Neurosci. 18, 965–977.

Evans, S.J., Bassis, C.M., Hein, R., Assari, S., Flowers, S.A., Kelly, M.B., McInnis, M.G., 2017. The gut microbiome composition associates with bipolar disorder and illness severity. J. Psychiatr. Res. 87, 23–29.

Fan, F., Lei, M., 2022. Mechanisms Underlying Curcumin-Induced Neuroprotection in Cerebral Ischemia. Front. Pharmacol. 13, 893118 https://doi.org/10.3389/fphar.2022.893118.

Fan, Y., Støving, R.K., Berreira Ibraim, S., Hyötyläinen, T., Thirion, F., Arora, T., Lyu, L., Stankevic, E., Hansen, T.H., Déchelotte, P., 2023. The gut microbiota contributes to the pathogenesis of anorexia nervosa in humans and mice. Nat. Microbiol. 1–16.
Fava, F., Rizzetto, L., Tuohy, K.M., 2019. Gut microbiota and health: Connecting actors across the metabolic system. Proc. Nutr. Soc. 78, 177–188.
Felger, J.C., Lotrich, F.E., 2013. Inflammatory cytokines in depression: neurobiological mechanisms and therapeutic implications. Neuroscience 246, 199–229.
Fitton, R., Sweetman, J., Heseltine-Carp, W., van der Feltz-Cornelis, C., 2022. Anti-inflammatory medications for the treatment of mental disorders: A scoping review. Brain Behav. Immun. Health 26, 100518. https://doi.org/10.1016/j.bbih.2022.100518.
Flowers, S.A., Ward, K.M., Clark, C.T., 2020. The gut microbiome in bipolar disorder and pharmacotherapy management. Neuropsychobiology 79 (1-2), 43–49.
Fornaro, M., Anastasia, A., Monaco, F., Novello, S., Fusco, A., Iasevoli, F., De Berardis, D., Veronese, N., Solmi, M., de Bartolomeis, A., 2018. Clinical and psychopathological features associated with treatment-emergent mania in bipolar-II depressed outpatients exposed to antidepressants. J. Affect. Disord. 234, 131–138.
Forssberg, H., 2019. Microbiome programming of brain development: implications for neurodevelopmental disorders. Dev. Med. Child Neurol. 61, 744–749.
Fowlie, G., Cohen, N., Ming, X., 2018. The perturbance of microbiome and gut-brain axis in autism spectrum disorders. Int. J. Mol. Sci. 19 (8), 2251.
Fröhlich, E.E., Farzi, A., Mayerhofer, R., Reichmann, F., Jačan, A., Wagner, B., Zinser, E., Bordag, N., Magnes, C., Fröhlich, E., Kashofer, K., Gorkiewicz, G., Holzer, P., 2016. Cognitive impairment by antibiotic-induced gut dysbiosis: Analysis of gut microbiota-brain communication. Brain Behav. Immun. 56, 140–155. https://doi.org/10.1016/j.bbi.2016.02.020.
Frye, R.E., Rose, S., Chacko, J., Wynne, R., Bennuri, S.C., Slattery, J.C., Tippett, M., Delhey, L., Melnyk, S., Kahler, S.G., 2016. Modulation of mitochondrial function by the microbiome metabolite propionic acid in autism and control cell lines. Transl. Psychiatry 6 e927–e927.
Fujii, Y., Khasnobish, A., Morita, H., 2019. Relationship between Alzheimer's Disease and the Human Microbiome. In: Wisniewski, T. (Ed.), Alzheimer's Disease. Codon Publications, Brisbane (AU).
Fukui, H., 2016. Increased intestinal permeability and decreased barrier function: does it really influence the risk of inflammation? Inflamm. Intest. Dis. 1, 135–145.
Gantz, I., Fong, T.M., 2003. The melanocortin system. Am. J. Physiol. -Endocrinol. Metab. 284, E468–E474. https://doi.org/10.1152/ajpendo.00434.2002.
Gao, M., Wang, J., Liu, P., Tu, H., Zhang, R., Zhang, Y., Zhang, K., 2023. Gut microbiota composition in depressive disorder: a systematic review, meta-analysis, and meta-regression. Transl. Psychiatry 13 (1), 379.
Garcia, N., Gutierrez, E., 2023. Anorexia nervosa and microbiota: systematic review and critical appraisal. Eating and Weight Disorders-Studies on Anorexia, Bulimia and Obesity, 28 (1), 1.
Ghosh, T.S., Das, M., Jeffery, I.B., O'Toole, P.W., 2020. Adjusting for age improves identification of gut microbiome alterations in multiple diseases. Elife 9, e50240.
Goehler, L.E., Gaykema, R.P., Opitz, N., Reddaway, R., Badr, N., Lyte, M., 2005. Activation in vagal afferents and central autonomic pathways: early responses to intestinal infection with Campylobacter jejuni. Brain, Behav., Immun. 19, 334–344.
Goehler, L.E., Park, S.M., Opitz, N., Lyte, M., Gaykema, R.P., 2008. Campylobacter jejuni infection increases anxiety-like behavior in the holeboard: possible anatomical substrates for viscerosensory modulation of exploratory behavior. Brain, Behav., Immun. 22, 354–366.
Góralczyk-Bińkowska, A., Szmajda-Krygier, D., Kozłowska, E., 2022. The microbiota–gut–brain Axis in psychiatric disorders. Int. J. Mol. Sci. 23, 11245.
Guinane, C.M., Cotter, P.D., 2013. Role of the gut microbiota in health and chronic gastrointestinal disease: understanding a hidden metabolic organ. Ther. Adv. Gastroenterol. 6, 295–308.
Gulati, M., Singh, S.K., Corrie, L., Kaur, I.P., Chandwani, L., 2020. Delivery routes for faecal microbiota transplants: Available, anticipated and aspired. Pharm. Res 159, 104954. https://doi.org/10.1016/j.phrs.2020.104954.
Gupta, S., Allen-Vercoe, E., Petrof, E.O., 2016. Fecal microbiota transplantation: in perspective. Ther. Adv. Gastroenterol. 9, 229–239. https://doi.org/10.1177/1756283×15607414.
Han, H., Jang, J., 2022. Recent advances in biofabricated gut models to understand the gut-brain axis in neurological diseases. Front. Med. Technol. 4, 931411.
Hasegawa, S., Goto, S., Tsuji, H., Okuno, T., Asahara, T., Nomoto, K., Shibata, A., Fujisawa, Y., Minato, T., Okamoto, A., Ohno, K., Hirayama, M., 2015. Intestinal dysbiosis and lowered serum lipopolysaccharide-binding protein in Parkinson's disease. PloS One 10 (11), e0142164.
He, J., Zhang, P., Shen, L., Niu, L., Tan, Y., Chen, L., Zhao, Y., Bai, L., Hao, X., Li, X., 2020. Short-chain fatty acids and their association with signalling pathways in inflammation, glucose and lipid metabolism. Int. J. Mol. Sci. 21, 6356.
Hinds, J.A., Sanchez, E.R., 2022. The role of the hypothalamus–pituitary–adrenal (HPA) axis in test-induced anxiety: assessments, physiological responses, and molecular details. Stresses 2, 146–155.
Ho, J., Nicolucci, A.C., Virtanen, H., Schick, A., Meddings, J., Reimer, R.A., Huang, C., 2019. Effect of Prebiotic on Microbiota, Intestinal Permeability, and Glycemic Control in Children With Type 1 Diabetes. J. Clin. Endocrinol. Metab. 104, 4427–4440. https://doi.org/10.1210/jc.2019-00481.
Hsiao, E.Y., McBride, S.W., Hsien, S., Sharon, G., Hyde, E.R., McCue, T., Codelli, J.A., Chow, J., Reisman, S.E., Petrosino, J.F., 2013. Microbiota modulate behavioral and physiological abnormalities associated with neurodevelopmental disorders. Cell 155, 1451–1463.
Iovene, M.R., Bombace, F., Maresca, R., Sapone, A., Iardino, P., Picardi, A., Marotta, R., Schiraldi, C., Siniscalco, D., Serra, N., 2017. Intestinal dysbiosis and yeast isolation in stool of subjects with autism spectrum disorders. Mycopathologia 182, 349–363.
Jiang, H., Zhou, Y., Zhou, G., Li, Y., Yuan, J., Li, X., Ruan, B., 2018. Gut microbiota profiles in treatment-naïve children with attention deficit hyperactivity disorder. Behav. Brain Res. 347, 408–413.
Johnson, J.S., Spakowicz, D.J., Hong, B.-Y., Petersen, L.M., Demkowicz, P., Chen, L., Leopold, S.R., Hanson, B.M., Agresta, H.O., Gerstein, M., Sodergren, E., Weinstock, G.M., 2019. Evaluation of 16S rRNA gene sequencing for species and strain-level microbiome analysis. Nat. Commun. 10, 5029. https://doi.org/10.1038/s41467-019-13036-1.
Johnson, D., Thurairajasingam, S., Letchumanan, V., Chan, K.-G., Lee, L.-H., 2021. Exploring the Role and Potential of Probiotics in the Field of Mental Health: Major Depressive Disorder. Nutrients 13, 1728. https://doi.org/10.3390/nu13051728.
Kalenik, A., Kardaś, K., Rahnama, A., Sirojć, K., Wolańczyk, T., 2021. Gut microbiota and probiotic therapy in ADHD: A review of current knowledge. Prog. Neuro-Psychopharmacol. Biol. Psychiatry 110, 110277.
Katzenschlager, R., Evans, A., Manson, A., Patsalos, P.N., Ratnaraj, N., Watt, H., Timmermann, L., Van der Giessen, R., Lees, A.J., 2004. Mucuna pruriens in Parkinson's disease: a double blind clinical and pharmacological study. J. Neurol. Neurosurg. Psychiatry 75, 1672–1677. https://doi.org/10.1136/jnnp.2003.028761.
Keshavarzian, A., Green, S.J., Engen, P.A., Voigt, R.M., Naqib, A., Forsyth, C.B., Mutlu, E., Shannon, K.M., 2015. Colonic bacterial composition in Parkinson's disease. Mov. Disord. 30, 1351–1360. https://doi.org/10.1002/mds.26307.
Khayatan, D., Razavi, S.M., Arab, Z.N., Niknejad, A.H., Nouri, K., Momtaz, S., Gumpricht, E., Jamialahmadi, T., Abdolghaffari, A.H., Barreto, G.E., Sahebkar, A., 2022. Protective effects of curcumin against traumatic brain injury. Biomed. Pharmacother. 154, 113621 https://doi.org/10.1016/j.biopha.2022.113621.
Kim, Y.K., Shin, C., 2018. The microbiota-gut-brain axis in neuropsychiatric disorders: pathophysiological mechanisms and novel treatments. Curr. Neuropharmacol. 16 (5), 559–573.
Kong, G.Y.-E., Letchumanan, V., Tan, L.T.-H., Law, J.W.-F., 2022. Gut Microbiome in Obsessive Compulsive Disorder: Potential of Probiotics as an Adjuvant Therapy. Prog. Microbes Mol. Biol. 5.
Kraimi, N., Lormant, F., Calandreau, L., Kempf, F., Zemb, O., Lemarchand, J., Constantin, P., Parias, C., Germain, K., Rabot, S., 2022. Microbiota and stress: a loop that impacts memory. Psychoneuroendocrinology 136, 105594.
Lampariello, L.R., Cortelazzo, A., Guerranti, R., Sticozzi, C., Valacchi, G., 2012. The Magic Velvet Bean of Mucuna pruriens. J. Tradit. Complement. Med. 2, 331–339. https://doi.org/10.1016/S2225-4110(16)30119-5.
Lapidot, Y., Reshef, L., Maya, M., Cohen, D., Gophna, U., Muhsen, K., 2022. Socioeconomic disparities and household crowding in association with the fecal microbiome of school-age children. npj Biofilms and Microbiomes 8 (1), 10.
Leclercq, S., Matamoros, S., Cani, P.D., Neyrinck, A.M., Jamar, F., Stärkel, P., Windey, K., Tremaroli, V., Bäckhed, F., Verbeke, K., 2014. Intestinal permeability, gut-bacterial dysbiosis, and behavioral markers of alcohol-dependence severity. Proc. Natl. Acad. Sci. 111, E4485–E4493.
Li, J., Zhao, F., Wang, Y., Chen, J., Tao, J., Tian, G., Wu, S., Liu, W., Cui, Q., Geng, B., Zhang, W., Weldon, R., Auguste, K., Yang, L., Liu, X., Chen, L., Yang, X., Zhu, B., Cai, J., 2017. Gut microbiota dysbiosis contributes to the development of hypertension. Microbiome 5 (1), 14.
Li, Q., Zhou, J.-M., 2016. The microbiota–gut–brain axis and its potential therapeutic role in autism spectrum disorder. Neuroscience 324, 131–139.
Lin, A., Zheng, W., He, Y., Tang, W., Wei, X., He, R., Huang, W., Su, Y., Huang, Y., Zhou, H., Xie, H., 2018. Gut microbiota in patients with Parkinson's disease in southern China. Park. Relat. Disord. 53, 82–88. https://doi.org/10.1016/j.parkreldis.2018.05.007.
Liu, L., Wang, H., Chen, X., Zhang, Y., Zhang, H., Xie, P., 2023. Gut microbiota and its metabolites in depression: from pathogenesis to treatment. EBioMedicine 90.
Liu, B., Ye, D., Yang, H., Song, J., Sun, X., Mao, Y., He, Z., 2022. Two-sample mendelian randomization analysis investigates causal associations between gut microbial genera and inflammatory bowel disease, and specificity causal associations in ulcerative colitis or Crohn's disease. Front. Immunol. 13, 921546.
Lopera-Maya, E.A., Kurilshikov, A., van der Graaf, A., Hu, S., Andreu-Sánchez, S., Chen, L., et al., 2022. Effect of host genetics on the gut microbiome in 7,738 participants of the Dutch Microbiome Project. Nature genetics 54 (2), 143–151.
Lormant, F., Ferreira, V.H.B., Meurisse, M., Lemarchand, J., Constantin, P., Morisse, M., Cornilleau, F., Parias, C., Chaillou, E., Bertin, A., 2020. Emotionality modulates the impact of chronic stress on memory and neurogenesis in birds. Sci. Rep. 10, 14620.
Lowe, V.M., Chaplin, M., Sgambato, D., 2023. Major depressive disorder and the gut microbiome: what is the link? Gen. Psychiatry 36 (1).
Ma, X., Asif, H., Dai, L., He, Y., Zheng, W., Wang, D., Ren, H., Tang, J., Li, C., Jin, K., 2020. Alteration of the gut microbiome in first-episode drug-naïve and chronic medicated schizophrenia correlate with regional brain volumes. J. Psychiatr. Res. 123, 136–144.
Ma, H., Cheng, N., Zhang, C., 2022. Schizophrenia and Alarmins. Medicina 58, 694.
Madison, A., Kiecolt-Glaser, J.K., 2019. Stress, depression, diet, and the gut microbiota: human–bacteria interactions at the core of psychoneuroimmunology and nutrition. Curr. Opin. Behav. Sci. 28, 105–110.
Magistrelli, L., Amoruso, A., Mogna, L., Graziano, T., Cantello, R., Pane, M., Comi, C., 2019. Probiotics May Have Beneficial Effects in Parkinson's Disease: In vitro Evidence. Front Immunol. 10, 969. https://doi.org/10.3389/fimmu.2019.00969.
Malan-Muller, S., Valles-Colomer, M., Raes, J., Lowry, C.A., Seedat, S., Hemmings, S.M., 2018. The gut microbiome and mental health: implications for anxiety-and trauma-related disorders. Omics: a J. Integr. Biol. 22, 90–107.
Manach, C., Scalbert, A., Morand, C., Rémésy, C., Jiménez, L., 2004. Polyphenols: food sources and bioavailability. Am. J. Clin. Nutr. 79, 727–747. https://doi.org/10.1093/ajcn/79.5.727.

Mandel, S., Weinreb, O., Amit, T., Youdim, M.B.H., 2004. Cell signaling pathways in the neuroprotective actions of the green tea polyphenol (-)-epigallocatechin-3-gallate: implications for neurodegenerative diseases. J. Neurochem 88, 1555–1569. https://doi.org/10.1046/j.1471-4159.2003.02291.x.

Mao, J.-H., Kim, Y.-M., Zhou, Y.-X., Hu, D., Zhong, C., Chang, H., Brislawn, C.J., Fansler, S., Langley, S., Wang, Y., Peisl, B.Y.L., Celniker, S.E., Threadgill, D.W., Wilmes, P., Orr, G., Metz, T.O., Jansson, J.K., Snijders, A.M., 2020. Genetic and metabolic links between the murine microbiome and memory. Microbiome 8, 53. https://doi.org/10.1186/s40168-020-00817-w.

Marano, G., Mazza, M., Lisci, F.M., Ciliberto, M., Traversi, G., Kotzalidis, G.D., De Berardis, D., Laterza, L., Sani, G., Gasbarrini, A., 2023. The Microbiota–Gut–Brain Axis: Psychoneuroimmunological Insights. Nutrients 15, 1496.

Marler, S., Ferguson, B.J., Lee, E.B., Peters, B., Williams, K.C., McDonnell, E., Macklin, E. A., Levitt, P., Gillespie, C.H., Anderson, G.M., 2016. Brief report: whole blood serotonin levels and gastrointestinal symptoms in autism spectrum disorder. J. Autism Dev. Disord. 46, 1124–1130.

Marques, T.M., Cryan, J.F., Shanahan, F., Fitzgerald, G.F., Ross, R.P., Dinan, T.G., Stanton, C., 2014. Gut microbiota modulation and implications for host health: Dietary strategies to influence the gut–brain axis. Innov. Food Sci. Emerg. Technol. 22, 239–247.

Martin, M.J., Lara-Villoslada, F., Ruiz, M.A., Morales, M.E., 2013. Effect of unmodified starch on viability of alginate-encapsulated Lactobacillus fermentum CECT5716. LWT - Food Sci. Technol. 53, 480–486. https://doi.org/10.1016/j.lwt.2013.03.019.

Mass, M., Kubera, M., Leunis, J.-C., 2008. The gut-brain barrier in major depression: intestinal mucosal dysfunction with an increased translocation of LPS from gram negative enterobacteria (leaky gut) plays a role in the inflammatory pathophysiology of depression. Neuroendocrinol. Lett. 29, 117–124.

Mattheisen, M., Samuels, J.F., Wang, Y., Greenberg, B.D., Fyer, A.J., McCracken, J.T., Geller, D.A., Murphy, D.L., Knowles, J.A., Grados, M.A., 2015. Genome-wide association study in obsessive-compulsive disorder: results from the OCGAS. Mol. Psychiatry 20, 337–344.

Mayer, E.A., 2011. Gut feelings: the emerging biology of gut–brain communication. Nat. Rev. Neurosci. 12, 453–466.

McGuinness, A.J., Davis, J.A., Dawson, S.L., Loughman, A., Collier, F., O'hely, M., Jacka, F.N., 2022. A systematic review of gut microbiota composition in observational studies of major depressive disorder, bipolar disorder and schizophrenia. Mol. Psychiatry 27 (4), 1920–1935.

McIntyre, R.S., Subramaniapillai, M., Shekotikhina, M., Carmona, N.E., Lee, Y., Mansur, R.B., Brietzke, E., Fus, D., Coles, A.S., Iacobucci, M., 2021. Characterizing the gut microbiota in adults with bipolar disorder: a pilot study. Nutr. Neurosci. 24, 173–180.

Messaoudi, M., Violle, N., Bisson, J.-F., Desor, D., Javelot, H., Rougeot, C., 2011. Beneficial psychological effects of a probiotic formulation (Lactobacillus helveticus R0052 and Bifidobacterium longum R0175) in healthy human volunteers. Gut Microbes 2, 256–261. https://doi.org/10.4161/gmic.2.4.16108.

Micioni Di Bonaventura, E., Botticelli, L., Tomassoni, D., Tayebati, S.K., Micioni Di Bonaventura, M.V., Cifani, C., 2020. The Melanocortin System behind the Dysfunctional Eating Behaviors. Nutrients 12, 3502. https://doi.org/10.3390/nu12113502.

Mikulska, J., Juszczyk, G., Gawrońska-Grzywacz, M., Herbet, M., 2021. HPA axis in the pathomechanism of depression and schizophrenia: new therapeutic strategies based on its participation. Brain Sci. 11, 1298.

Miyake, S., Kim, S., Suda, W., Oshima, K., Nakamura, M., Matsuoka, T., Chihara, N., Tomita, A., Sato, W., Kim, S.W., Morita, H., Hattori, M., Yamamura, T., 2015. Dysbiosis in the Gut Microbiota of Patients with Multiple Sclerosis, with a Striking Depletion of Species Belonging to Clostridia XIVa and IV Clusters. PloS One 10 (9), e0137429.

Miyaoka, T., Wake, R., Furuya, M., Liaury, K., Ieda, M., Kawakami, K., Tsuchie, K., Taki, M., Ishihara, K., Araki, T., 2012. Minocycline as adjunctive therapy for patients with unipolar psychotic depression: an open-label study. Prog. neuro-Psychopharmacol. Biol. Psychiatry 37, 222–226.

Morais, L.H., Schreiber IV, H.L., Mazmanian, S.K., 2021. The gut microbiota–brain axis in behaviour and brain disorders. Nat. Rev. Microbiol. 19, 241–255.

Moreno-Navarrete, J.M., Sabater, M., Ortega, F., Ricart, W., Fernandez-Real, J.M., 2012. Circulating zonulin, a marker of intestinal permeability, is increased in association with obesity-associated insulin resistance. PloS One 7 (5), e37160.

Morgan, X.C., Huttenhower, C., 2012. Chapter 12: Human microbiome analysis. PLoS Comput. Biol. 8, e1002808.

Mörkl, S., Butler, M.I., Holl, A., Cryan, J.F., Dinan, T.G., 2020. Probiotics and the Microbiota-Gut-Brain Axis: Focus on Psychiatry. Curr. Nutr. Rep. 9, 171–182. https://doi.org/10.1007/s13668-020-00313-5.

Morris, G., Fernandes, B.S., Puri, B.K., Walker, A.J., Carvalho, A.F., Berk, M., 2018. Leaky brain in neurological and psychiatric disorders: Drivers and consequences. Aust. N. Z. J. Psychiatry 52, 924–948.

Moysidou, C.-M., Owens, R.M., 2021. Advances in modelling the human microbiome–gut–brain axis in vitro. Biochem. Soc. Trans. 49, 187–201.

Müller, N., 2017. Immunological aspects of the treatment of depression and schizophrenia. Dialog-. Clin. Neurosci. 19, 55–63. https://doi.org/10.31887/DCNS.2017.19.1/nmueller.

Munkholm, K., Braüner, J.V., Kessing, L.V., Vinberg, M., 2013. Cytokines in bipolar disorder vs. healthy control subjects: a systematic review and meta-analysis. J. Psychiatr. Res. 47, 1119–1133.

Murrough, J.W., Stade, E., Sayed, S., Ahle, G., Kiraly, D.D., Welch, A., Collins, K.A., Soleimani, L., Iosifescu, D.V., Charney, D.S., 2017. Dextromethorphan/quinidine pharmacotherapy in patients with treatment resistant depression: a proof of concept clinical trial. J. Affect. Disord. 218, 277–283.

Nezi, M., Mastorakos, G., Mouslech, Z., 2015. Corticotropin releasing hormone and the immune/inflammatory response.

Nguyen, T.T., Eyler, L.T., Jeste, D.V., 2018. Systemic biomarkers of accelerated aging in schizophrenia: a critical review and future directions. Schizophr. Bull. 44, 398–408.

Nguyen, T.T., Hathaway, H., Kosciolek, T., Knight, R., Jeste, D.V., 2021. Gut microbiome in serious mental illnesses: A systematic review and critical evaluation. Schizophr. Res. 234, 24–40.

Ni, J.-J., Xu, Q., Yan, S.-S., Han, B.-X., Zhang, H., Wei, X.-T., Feng, G.-J., Zhao, M., Pei, Y.-F., Zhang, L., 2022. Gut microbiota and psychiatric disorders: a two-sample Mendelian randomization study. Front. Microbiol. 12, 737197.

Nikolova, V.L., Smith, M.R., Hall, L.J., Cleare, A.J., Stone, J.M., Young, A.H., 2021. Perturbations in gut microbiota composition in psychiatric disorders: a review and meta-analysis. JAMA Psychiatry 78, 1343–1354.

Nocera, A., Nasrallah, H., 2022. The Association of the Gut Microbiota with Clinical Features in Schizophrenia. Behav. Sci. 12, 89. https://doi.org/10.3390/bs12040089.

O'Brien, S.M., Scully, P., Scott, L.V., Dinan, T.G., 2006. Cytokine profiles in bipolar affective disorder: focus on acutely ill patients. J. Affect. Disord. 90, 263–267.

O'Mahony, S.M., Clarke, G., Borre, Y.E., Dinan, T.G., Cryan, J.F., 2015. Serotonin, tryptophan metabolism and the brain-gut-microbiome axis. Behav. brain Res. 277, 32–48.

Odamaki, T., Kato, K., Sugahara, H., Hashikura, N., Takahashi, S., Xiao, J.Z., Osawa, R., 2016. Age-related changes in gut microbiota composition from newborn to centenarian: a cross-sectional study. BMC microbiology 16 (1), 1–12.

Ogunrinola, G.A., Oyewale, J.O., Oshamika, O.O., Olasehinde, G.I., 2020. The Human Microbiome and Its Impacts on Health. Int. J. Microbiol. 2020, 1–7. https://doi.org/10.1155/2020/8045646.

Ono, E., Hatayama, M., Isono, Y., Sato, T., Watanabe, R., Yonekura-Sakakibara, K., Fukuchi-Mizutani, M., Tanaka, Y., Kusumi, T., Nishino, T., Nakayama, T., 2006. Localization of a flavonoid biosynthetic polyphenol oxidase in vacuoles. Plant J. 45, 133–143. https://doi.org/10.1111/j.1365-313X.2005.02625.x.

Ortega, M.A., Álvarez-Mon, M.A., García-Montero, C., Fraile-Martínez, Ó., Monserrat, J., Martinez-Rozas, L., Rodríguez-Jiménez, R., Álvarez-Mon, M., Lahera, G., 2023. Microbiota–gut–brain axis mechanisms in the complex network of bipolar disorders: potential clinical implications and translational opportunities. Mol. Psychiatry 1–29.

Ortiz-Dominguez, A., Berlanga, C., Gutierrez, D., Hernandez, M., Pavon, L., 2005. Neuroimmune interactions in bipolar disorder: phasic differences, in: BIPOLAR DISORDERS. BLACKWELL PUBLISHING 9600 GARSINGTON RD, OXFORD OX4 2DQ, OXON, ENGLAND, pp. 83–83.

Osadchiy, V., Martin, C.R., Mayer, E.A., 2011. Gut microbiome and modulation of CNS function. Comprehensive. Physiology 10, 57–72.

Parker, G., Spoelma, M.J., Rhodes, N., 2022. Faecal microbiota transplantation for bipolar disorder: A detailed case study. Bipolar Disord. 24, 559–563. https://doi.org/10.1111/bdi.13187.

Pascal, M., Perez-Gordo, M., Caballero, T., Escribese, M.M., Lopez Longo, M.N., Luengo, O., Manso, L., Matheu, V., Seoane, E., Zamorano, M., 2018. Microbiome and allergic diseases. Front. Immunol. 9, 1584.

Patrascu, O., Béguet-Crespel, F., Marinelli, L., Le Chatelier, E., Abraham, A.-L., Leclerc, M., Klopp, C., Terrapon, N., Henrissat, B., Blottière, H.M., 2017. A fibrolytic potential in the human ileum mucosal microbiota revealed by functional metagenomic. Sci. Rep. 7, 40248.

Patterson, E., Ryan, P.M., Wiley, N., Carafa, I., Sherwin, E., Moloney, G., Franciosi, E., Mandal, R., Wishart, D.S., Tuohy, K., 2019. Gamma-aminobutyric acid-producing lactobacilli positively affect metabolism and depressive-like behaviour in a mouse model of metabolic syndrome. Sci. Rep. 9, 16323.

Peterson, C.T., Sharma, V., Elmén, L., Peterson, S.N., 2015. Immune homeostasis, dysbiosis and therapeutic modulation of the gut microbiota. Clin. Exp. Immunol. 179, 363–377. https://doi.org/10.1111/cei.12474.

Petschow, B., Doré, J., Hibberd, P., Dinan, T., Reid, G., Blaser, M., Cani, P.D., Degnan, F. H., Foster, J., Gibson, G., Hutton, J., Klaenhammer, T.R., Ley, R., Nieuwdorp, M., Pot, B., Relman, D., Serazin, A., Sanders, M.E., 2013. Probiotics, prebiotics, and the host microbiome: the science of translation. Ann. N. Y Acad. Sci. 1306, 1–17. https://doi.org/10.1111/nyas.12303.

Phavichitr, N., Wang, S., Chomto, S., Tantibhaedhyangkul, R., Kakourou, A., Intarakhao, S., Knol, J., 2021. Impact of synbiotics on gut microbiota during early life: A randomized, double-blind study. Sci. Rep. 11 (1), 3534.

Philips, C.A., Augustine, P., Ganesan, K., Ranade, S., Chopra, V., Patil, K., Shende, S., Ahamed, R., Kumbar, S., Rajesh, S., 2022. The role of gut microbiota in clinical complications, disease severity, and treatment response in severe alcoholic hepatitis. Indian J. Gastroenterol. 41, 37–51.

Piccioni, A., Cicchinelli, S., Valletta, F., De Luca, G., Longhitano, Y., Candelli, M., Ojetti, V., Sardeo, F., Navarra, S., Covino, M., 2022. Gut microbiota and autoimmune diseases: a charming real world together with probiotics. Curr. Med. Chem. 29, 3147–3159.

Poretsky, R., Rodriguez-R, L.M., Luo, C., Tsementzi, D., Konstantinidis, K.T., 2014. Strengths and limitations of 16S rRNA gene amplicon sequencing in revealing temporal microbial community dynamics. PloS One 9 (4), e93827.

Portincasa, P., Bonfrate, L., Vacca, M., De Angelis, M., Farella, I., Lanza, E., Khalil, M., Wang, D.Q.-H., Sperandio, M., Di Ciaula, A., 2022. Gut Microbiota and Short Chain Fatty Acids: Implications in Glucose Homeostasis. IJMS 23, 1105. https://doi.org/10.3390/ijms23031105.

Putignani, L., Gasbarrini, A., Dallapiccola, B., 2019. Potential of multiomics technology in precision medicine. Curr. Opin. Gastroenterol. 35, 491.

Ramamoorthy, S., Cidlowski, J.A., 2016. Corticosteroids: mechanisms of action in health and disease. Rheum. Dis. Clin. 42, 15–31.

Rezai-Zadeh, K., Shytle, D., Sun, N., Mori, T., Hou, H., Jeanniton, D., Ehrhart, J., Townsend, K., Zeng, J., Morgan, D., Hardy, J., Town, T., Tan, J., 2005. Green tea

epigallocatechin-3-gallate (EGCG) modulates amyloid precursor protein cleavage and reduces cerebral amyloidosis in Alzheimer transgenic mice. J. Neurosci. 25, 8807–8814. https://doi.org/10.1523/JNEUROSCI.1521-05.2005.
Richarte, V., Sánchez-Mora, C., Corrales, M., Fadeuilhe, C., Vilar-Ribó, L., Arribas, L., Garcia, E., Rosales-Ortiz, S.K., Arias-Vasquez, A., Soler-Artigas, M., 2021. Gut microbiota signature in treatment-naïve attention-deficit/hyperactivity disorder. Transl. Psychiatry 11, 382.
Roager, H.M., Licht, T.R., 2018. Microbial tryptophan catabolites in health and disease. Nat. Commun. 9, 3294.
Rumah, K.R., Linden, J., Fischetti, V.A., Vartanian, T., 2013. Isolation of Clostridium perfringens type B in an individual at first clinical presentation of multiple sclerosis provides clues for environmental triggers of the disease. PloS One 8 (10), e76359.
Rutsch, A., Kantsjö, J.B., Ronchi, F., 2020. The gut-brain axis: how microbiota and host inflammasome influence brain physiology and pathology. Front. Immunol. 11, 604179.
Santoro, A., Ostan, R., Candela, M., Biagi, E., Brigidi, P., Capri, M., Franceschi, C., 2018. Gut microbiota changes in the extreme decades of human life: a focus on centenarians. Cell. Mol. Life Sci. 75, 129–148. https://doi.org/10.1007/s00018-017-2674-y.
Scheperjans, F., Aho, V., Pereira, P.A.B., Koskinen, K., Paulin, L., Pekkonen, E., Haapaniemi, E., Kaakkola, S., Eerola-Rautio, J., Pohja, M., Kinnunen, E., Murros, K., Auvinen, P., 2015. Gut microbiota are related to Parkinson's disease and clinical phenotype. Mov. Disord. 30, 350–358. https://doi.org/10.1002/mds.26069.
Schiele, M.A., Thiel, C., Deckert, J., Zaudig, M., Berberich, G., Domschke, K., 2020. Monoamine oxidase A hypomethylation in obsessive-compulsive disorder: reversibility by successful psychotherapy? Int. J. Neuropsychopharmacol. 23, 319–323.
Schleupner, H.V., Carmichael, M.J., 2022. Attention-Deficit/Hyperactivity Disorder and the gut microbiota–gut–brain axis: Closing research gaps through female inclusion in study design. Women 2, 231–253.
Schwarz, E., Maukonen, J., Hyytiäinen, T., Kieseppä, T., Orešič, M., Sabunciyan, S., Mantere, O., Saarela, M., Yolken, R., Suvisaari, J., 2018. Analysis of microbiota in first episode psychosis identifies preliminary associations with symptom severity and treatment response. Schizophr. Res. 192, 398–403.
Seetharaman, S., Fields, E.L., 2020. Avoidant/Restrictive Food Intake Disorder. Pediatr. Rev. 41, 613–622. https://doi.org/10.1542/pir.2019-0133.
Sender, R., Fuchs, S., Milo, R., 2016. Revised estimates for the number of human and bacteria cells in the body. PLoS Biol. 14, e1002533.
Sergeev, I.N., Aljutaily, T., Walton, G., Huarte, E., 2020. Effects of synbiotic supplement on human gut microbiota, body composition and weight loss in obesity. Nutrients 12 (1), 222.
Shen, Y., Xu, J., Li, Z., Huang, Y., Yuan, Y., Wang, J., Zhang, M., Hu, S., Liang, Y., 2018. Analysis of gut microbiota diversity and auxiliary diagnosis as a biomarker in patients with schizophrenia: a cross-sectional study. Schizophr. Res. 197, 470–477.
Shil, A., Chichger, H., 2021. Artificial sweeteners negatively regulate pathogenic characteristics of two model gut bacteria, E. coli and E. faecalis. International journal of molecular sciences 22 (10), 5228.
Shimizu, K., Yamada, T., Ogura, H., Mohri, T., Kiguchi, T., Fujimi, S., Shimazu, T., 2018. Synbiotics modulate gut microbiota and reduce enteritis and ventilator-associated pneumonia in patients with sepsis: a randomized controlled trial. Crit. Care 22, 1–9.
Silva, Y.P., Bernardi, A., Frozza, R.L., 2020. The role of short-chain fatty acids from gut microbiota in gut-brain communication. Front. Endocrinol. 11, 25.
Simpson, C.A., Diaz-Arteche, C., Eliby, D., Schwartz, O.S., Simmons, J.G., Cowan, C.S., 2021. The gut microbiota in anxiety and depression–A systematic review. Clin. Psychol. Rev. 83, 101943.
Singh, R.K., Chang, H.W., Yan, D.I., Lee, K.M., Ucmak, D., Wong, K., et al., 2017. Influence of diet on the gut microbiome and implications for human health. Journal of translational medicine 15 (1), 1–17.
Singh, P.K., Kotia, V., Ghosh, D., Mohite, G.M., Kumar, A., Maji, S.K., 2013. Curcumin Modulates α-Synuclein Aggregation and Toxicity. ACS Chem. Neurosci. 4, 393–407. https://doi.org/10.1021/cn3001203.
Singhal, G., Jaehne, E.J., Corrigan, F., Baune, B.T., 2014. Cellular and molecular mechanisms of immunomodulation in the brain through environmental enrichment. Front. Cell. Neurosci. 8, 97.
Sittipo, P., Choi, J., Lee, S., Lee, Y.K., 2022. The function of gut microbiota in immune-related neurological disorders: A review. J. Neuroinflamm. 19, 1–17.
Smedlund, K.B., Sanchez, E.R., Hinds, T.D., 2021. FKBP51 and the molecular chaperoning of metabolism. Trends Endocrinol. Metab. 32, 862–874.
Srikantha, P., Mohajeri, M.H., 2019. The possible role of the microbiota-gut-brain-axis in autism spectrum disorder. Int. J. Mol. Sci. 20 (9), 2115.
Staley, C., Hamilton, M.J., Vaughn, B.P., Graiziger, C.T., Newman, K.M., Kabage, A.J., Sadowsky, M.J., Khoruts, A., 2017. Successful Resolution of Recurrent Clostridium difficile Infection using Freeze-Dried, Encapsulated Fecal Microbiota; Pragmatic Cohort Study. Am. J. Gastroenterol. 112, 940–947. https://doi.org/10.1038/ajg.2017.6.
Steardo, L., Luciano, M., Sampogna, G., Zinno, F., Saviano, P., Staltari, F., Segura Garcia, C., De Fazio, P., Fiorillo, A., 2020. Efficacy of the interpersonal and social rhythm therapy (IPSRT) in patients with bipolar disorder: results from a real-world, controlled trial. Ann. Gen. Psychiatry 19, 1–7.
Steiger, H., 2004. Eating disorders and the serotonin connection: state, trait and developmental effects. J. Psychiatry Neurosci. 29, 20–29.
Strandwitz, P., Kim, K.H., Terekhova, D., Liu, J.K., Sharma, A., Levering, J., Lewis, K., 2019. GABA-modulating bacteria of the human gut microbiota. Nat. Microbiol. 4 (3), 396–403.
Sublette, M.E., Cheung, S., Lieberman, E., Hu, S., Mann, J.J., Uhlemann, A.C., Miller, J. M., 2021. Bipolar disorder and the gut microbiome: A systematic review. Bipolar Disord. 23 (6), 544–564.
Sudo, N., Chida, Y., Aiba, Y., Sonoda, J., Oyama, N., Yu, X.-N., Kubo, C., Koga, Y., 2004. Postnatal microbial colonization programs the hypothalamic-pituitary-adrenal system for stress response in mice. J. Physiol. 558, 263–275. https://doi.org/10.1113/jphysiol.2004.063388.
Sun, L.-J., Li, J.-N., Nie, Y.-Z., 2020. Gut hormones in microbiota-gut-brain cross-talk. Chin. Med. J. 133, 826–833.
Sun, L., Zhang, X., Zhang, Y., Zheng, K., Xiang, Q., Chen, N., He, Q., 2019. Antibiotic-induced disruption of gut microbiota alters local metabolomes and immune responses. Frontiers in cellular and infection microbiology 9, 99.
Sutherland, B.A., Shaw, O.M., Clarkson, A.N., Jackson, D.N., Sammut, I.A., Appleton, I., 2005. Neuroprotective effects of (-)-epigallocatechin gallate following hypoxia-ischemia-induced brain damage: novel mechanisms of action. FASEB J. 19, 258–260. https://doi.org/10.1096/fj.04-2806fje.
Tamtaji, O.R., Heidari-Soureshjani, R., Mirhosseini, N., Kouchaki, E., Bahmani, F., Aghadavod, E., Tajabadi-Ebrahimi, M., Asemi, Z., 2019. Probiotic and selenium co-supplementation, and the effects on clinical, metabolic and genetic status in Alzheimer's disease: A randomized, double-blind, controlled trial. Clin. Nutr. 38, 2569–2575. https://doi.org/10.1016/j.clnu.2018.11.034.
Tang, W., Zhu, H., Feng, Y., Guo, R., Wan, D., 2020. The impact of gut microbiota disorders on the blood–brain barrier. Infect. Drug Resist. 3351–3363.
Taniya, M.A., Chung, H.J., Al Mamun, A., Alam, S., Aziz, M.A., Emon, N.U., Xiao, J., 2022. Role of gut microbiome in autism spectrum disorder and its therapeutic regulation. Front. Cell. Infect. Microbiol. 12, 998.
Tatomir, A., Micu, C., Crivii, C., 2014. The impact of stress and glucocorticoids on memory. Clujul Med. 87, 3.
Terry, S.M., Barnett, J.A., Gibson, D.L., 2022. A critical analysis of eating disorders and the gut microbiome. J. Eat. Disord. 10, 154.
Thursby, E., Juge, N., 2017. Introduction to the human gut microbiota. Biochem. J. 474, 1823–1836.
Tian, H., Chen, Q., Yang, B., Qin, H., Li, N., 2021. Analysis of gut microbiome and metabolite characteristics in patients with slow transit constipation. Dig. Dis. Sci. 66, 3026–3035.
Ton, A.M.M., Campagnaro, B.P., Alves, G.A., Aires, R., Côco, L.Z., Arpini, C.M., Guerra E Oliveira, T., Campos-Toimil, M., Meyrelles, S.S., Pereira, T.M.C., Vasquez, E.C., 2020. Oxidative stress and Dementia in Alzheimer's Patients: effects of synbiotic supplementation. Oxid. Med Cell Longev. 2020, 2638703 https://doi.org/10.1155/2020/2638703.
Tremlett, H., Fadrosh, D.W., Faruqi, A.A., Zhu, F., Hart, J., Roalstad, S., Graves, J., Lynch, S., Waubant, E., US Network of Pediatric MS Centers, 2016. Gut microbiota in early pediatric multiple sclerosis: a case-control study. Eur. J. Neurol. 23 (8), 1308–1321.
Troyer, E.A., Kohn, J.N., Ecklu-Mensah, G., Aleti, G., Rosenberg, D.R., Hong, S., 2021. Searching for host immune-microbiome mechanisms in obsessive-compulsive disorder: A narrative literature review and future directions. Neurosci. Biobehav. Rev. 125, 517–534.
Tulkens, J., Vergauwen, G., Van Deun, J., Geeurickx, E., Dhondt, B., Lippens, L., Hendrix, A., 2020. Increased levels of systemic LPS-positive bacterial extracellular vesicles in patients with intestinal barrier dysfunction. Gut 69 (1), 191–193.
Turna, J., Grosman Kaplan, K., Anglin, R., Patterson, B., Soreni, N., Bercik, P., Surette, M. G., Van Ameringen, M., 2020. The gut microbiome and inflammation in obsessive-compulsive disorder patients compared to age-and sex-matched controls: a pilot study. Acta Psychiatr. Scand. 142, 337–347.
Unger, M.M., Spiegel, J., Dillmann, K.U., Grundmann, D., Philippeit, H., Bürmann, J., Faßbender, K., Schwiertz, A., Schäfer, K.H., 2016. Short chain fatty acids and gut microbiota differ between patients with Parkinson's disease and age-matched controls. Park. Relat. Disord. 32, 66–72.
Vakili, K., Fathi, M., Yaghoobpoor, S., Sayehmiri, F., Nazerian, Y., Nazerian, A., Mohamadkhani, A., Khodabakhsh, P., Réus, G.Z., Hajibeygi, R., 2022. The contribution of gut-brain axis to development of neurological symptoms in COVID-19 recovered patients: A hypothesis and review of literature. Front. Cell. Infect. Microbiol. 12, 983089.
Valles-Colomer, M., Falony, G., Darzi, Y., Tigchelaar, E.F., Wang, J., Tito, R.Y., Schiweck, C., Kurilshikov, A., Joossens, M., Wijmenga, C., 2019. The neuroactive potential of the human gut microbiota in quality of life and depression. Nat. Microbiol. 4, 623–632.
Vellingiri, B., Aishwarya, S.Y., Jancy, S.B., Abhishek, G.S., Vijayakumar, P., Narayanasamy, A., Mariappan, S., Sangeetha, R., Gopalakrishnan, A.V., Parthasarathi, R., 2022. An anxious relationship between Autism Spectrum Disorder and Gut Microbiota: A tangled chemistry? J. Clin. Neurosci. 99, 169–189.
Vuong, H.E., Pronovost, G.N., Williams, D.W., Coley, E.J., Siegler, E.L., Qiu, A., Kazantsev, M., Wilson, C.J., Rendon, T., Hsiao, E.Y., 2020. The maternal microbiome modulates fetal neurodevelopment in mice. Nature 586, 281–286.
Wang, L.-J., Yang, C.-Y., Chou, W.-J., Lee, M.-J., Chou, M.-C., Kuo, H.-C., Yeh, Y.-M., Lee, S.-Y., Huang, L.-H., Li, S.-C., 2020. Gut microbiota and dietary patterns in children with attention-deficit/hyperactivity disorder. Eur. Child Adolesc. Psychiatry 29, 287–297.
Wolkowitz, O.M., Epel, E.S., Reus, V.I., 2001. Stress hormone-related psychopathology: pathophysiological and treatment implications. World J. Biol. Psychiatry 2, 115–143.
Woodworth, H.L., Beekly, B.G., Batchelor, H.M., Bugescu, R., Perez-Bonilla, P., Schroeder, L.E., Leinninger, G.M., 2017. Lateral Hypothalamic Neurotensin Neurons Orchestrate Dual Weight Loss Behaviors via Distinct Mechanisms. Cell Rep. 21, 3116–3128. https://doi.org/10.1016/j.celrep.2017.11.068.

Xiao, L., Zhao, F., 2023. Microbial transmission, colonisation and succession: From pregnancy to infancy. Gut 72 (4), 772–786.

Xu, C., Zhu, H., Qiu, P., 2019. Aging progression of human gut microbiota. BMC microbiology 19, 1–10.

Yaguchi, K., Nishimura-Akiyoshi, S., Kuroki, S., Onodera, T., Itohara, S., 2014. Identification of transcriptional regulatory elements for Ntng1 and Ntng2 genes in mice. Mol. Brain 7 (1), 16.

Yang, C., Lin, X., Wang, X., Liu, H., Huang, J., Wang, S., 2022. The schizophrenia and gut microbiota: A bibliometric and visual analysis. Front. Psychiatry 13, 1022472. https://doi.org/10.3389/fpsyt.2022.1022472.

Yano, J.M., Yu, K., Donaldson, G.P., Shastri, G.G., Ann, P., Ma, L., Nagler, C.R., Ismagilov, R.F., Mazmanian, S.K., Hsiao, E.Y., 2015. Indigenous bacteria from the gut microbiota regulate host serotonin biosynthesis. Cell 161, 264–276.

Young, R.B., Marcelino, V.R., Chonwerawong, M., Gulliver, E.L., Forster, S.C., 2021. Key technologies for progressing discovery of microbiome-based medicines. Front. Microbiol. 12, 685935.

Yuan, X., Zhang, P., Wang, Y., Liu, Y., Li, X., Kumar, B.U., Hei, G., Lv, L., Huang, X.-F., Fan, X., 2018. Changes in metabolism and microbiota after 24-week risperidone treatment in drug naïve, normal weight patients with first episode schizophrenia. Schizophr. Res. 201, 299–306.

Zapała, B., Stefura, T., Wójcik-Pędziwiatr, M., Kabut, R., Bałajewicz-Nowak, M., Milewicz, T., Dudek, A., Stój, A., Rudzińska-Bar, M., 2021. Differences in the composition of gut microbiota between patients with Parkinson's disease and healthy controls: a cohort study. J. Clin. Med 10, 5698. https://doi.org/10.3390/jcm10235698.

Zhang, B., Chen, T., Cao, M., Yuan, C., Reiter, R.J., Zhao, Z., Zhao, Y., Chen, L., Fan, W., Wang, X., 2022a. Gut microbiota dysbiosis induced by decreasing endogenous melatonin mediates the pathogenesis of Alzheimer's disease and obesity. Front. Immunol. 13, 900132.

Zhang, P., Kong, L., Huang, H., Pan, Y., Zhang, D., Jiang, J., Shen, Y., Xi, C., Lai, J., Ng, C. H., 2022b. Gut microbiota–a potential contributor in the pathogenesis of bipolar disorder. Front. Neurosci. 16, 830748.

Zhang, J., Song, L., Wang, Y., Liu, C., Zhang, L., Zhu, S., Liu, S., Duan, L., 2019. Beneficial effect of butyrate-producing Lachnospiraceae on stress-induced visceral hypersensitivity in rats. J. Gastroenterol. Hepatol. 34, 1368–1376.

Zheng, P., Zeng, B., Liu, M., Chen, J., Pan, J., Han, Y., Liu, Y., Cheng, K., Zhou, C., Wang, H., 2019. The gut microbiome from patients with schizophrenia modulates the glutamate-glutamine-GABA cycle and schizophrenia-relevant behaviors in mice. Sci. Adv. 5, eaau8317.

Zheng, P., Zeng, B., Zhou, C., Liu, M., Fang, Z., Xu, X., Zeng, L., Chen, J., Fan, S., Du, X., 2016. Gut microbiome remodeling induces depressive-like behaviors through a pathway mediated by the host's metabolism. Mol. Psychiatry 21, 786–796.

Zhu, F., Guo, R., Wang, W., Ju, Y., Wang, Q., Ma, Q., Sun, Q., Fan, Y., Xie, Y., Yang, Z., 2020. Transplantation of microbiota from drug-free patients with schizophrenia causes schizophrenia-like abnormal behaviors and dysregulated kynurenine metabolism in mice. Mol. Psychiatry 25, 2905–2918.

Zhu, F., Tu, H., Chen, T., 2022. The microbiota–gut–brain axis in depression: the potential pathophysiological mechanisms and microbiota combined antidepression effect. Nutrients 14, 2081.

Zhuang, Z.Q., Shen, L.L., Li, W.W., Fu, X., Zeng, F., Gui, L., Lü, Y., Cai, M., Zhu, C., Tan, Y.L., Zheng, P., Li, H.Y., Zhu, J., Zhou, H.D., Bu, X.L., Wang, Y.J., 2018. Gut microbiota is altered in patients with Alzheimer's Disease. J. Alzheimer'S. Dis.: JAD 63 (4), 1337–1346.